

Projekt

testowanie i optymalizacja sieci CNN

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2. Model na sieci ResNet50 od zera (bez transfer learningu) na CPU

```
batch_size = 20
class_count = 4

img_height = 64
img_width = 64
```

Trening:

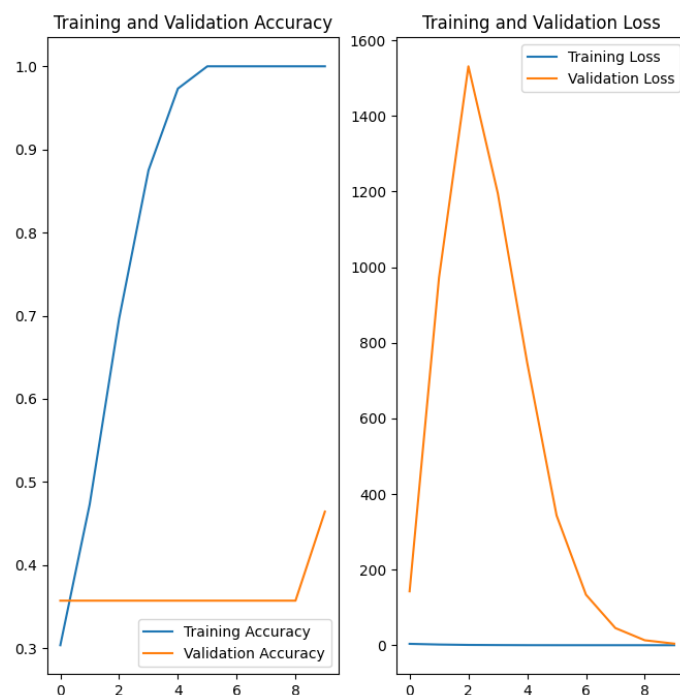
```
[63] epochs=10
     history = model.fit(
         train_ds,
         validation_data=val_ds,
         epochs=epochs
     )
```

Epoch 1/10
/usr/local/lib/python3.11/dist-packages/keras/src/backend/tensorflow/numpy.py:708: UserWarning: "`sparse\_categorical\_crossentropy` received an invalid argument for axis: 1" (Note that `axis=-1` refers to the last dimension, not the last axis in the shape tuple, which is different from the convention of NumPy).
output, from_logits = _get_logits(

Epoch	6/6	Time	Step	Accuracy	Loss	Val Accuracy	Val Loss
Epoch 1/10	6/6	70s	4s/step	0.3108	3.7121	0.3571	142.4979
Epoch 2/10	6/6	18s	3s/step	0.4263	2.1424	0.3571	971.8719
Epoch 3/10	6/6	17s	3s/step	0.7156	0.6699	0.3571	1531.4435
Epoch 4/10	6/6	17s	3s/step	0.8926	0.3293	0.3571	1195.6373
Epoch 5/10	6/6	18s	3s/step	0.9877	0.0674	0.3571	747.2027
Epoch 6/10	6/6	20s	3s/step	1.0000	0.0181	0.3571	343.1469
Epoch 7/10	6/6	18s	3s/step	1.0000	0.0075	0.3571	133.5202
Epoch 8/10	6/6	17s	3s/step	1.0000	0.0020	0.3571	45.3229
Epoch 9/10	6/6	20s	3s/step	1.0000	0.0014	0.3571	13.0383
Epoch 10/10	6/6	18s	3s/step	1.0000	5.5965e-04	0.4643	4.1082

Czas treningu wynosi dokładnie **3 minuty i 53 sekund**

Krzywe uczenia: dokładność i strata

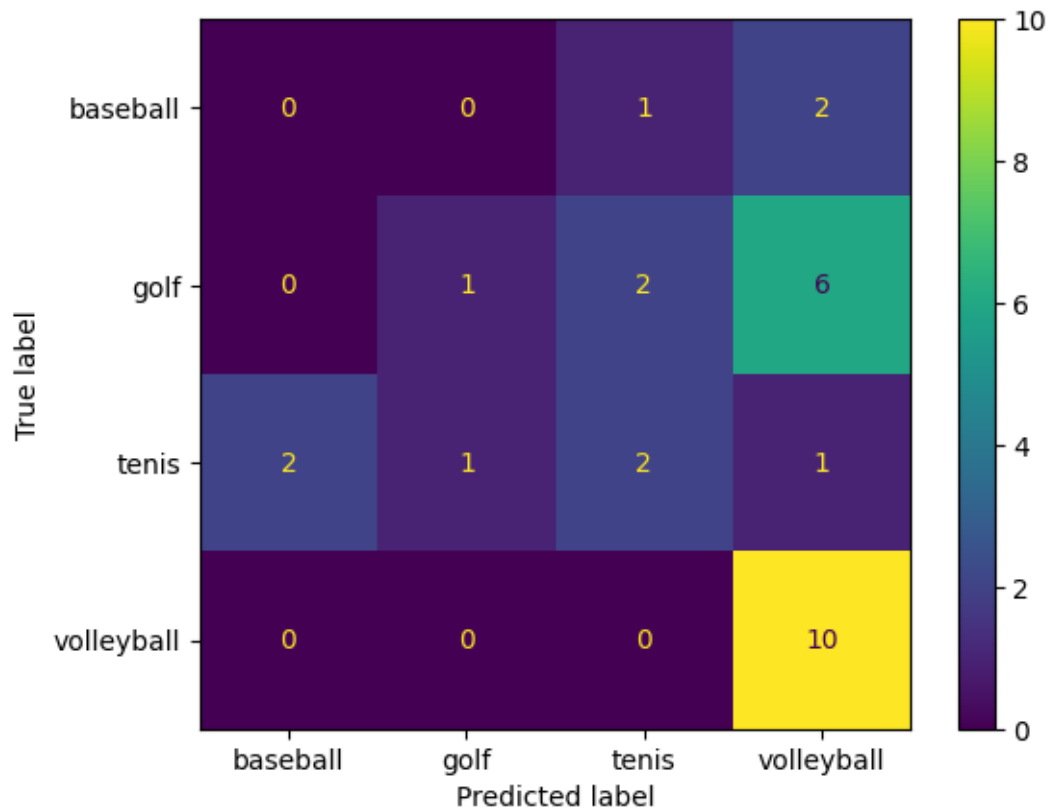


Analiza wyników:

	precision	recall	f1-score	support
baseball	0.00	0.00	0.00	3
golf	0.50	0.11	0.18	9
tenis	0.40	0.33	0.36	6
volleyball	0.53	1.00	0.69	10
accuracy			0.46	28
macro avg	0.36	0.36	0.31	28
weighted avg	0.43	0.46	0.38	28
[[0 0 1 2]				
[0 1 2 6]				
[2 1 2 1]				
[0 0 0 10]]				

Dokładność wynosi 0,46

Macierz błędów klasyfikacji



3a. Akcelerator GPU – raport porównawczy czas trenowania bez i z GPU

Czas treningu bez GPU wynosi dokładnie **3 minuty i 53 sekund**

```
[63] epochs=10
history = model.fit(
    train_ds,
    validation_data=val_ds,
    epochs=epochs
)
```

Epoch 1/10
/usr/local/lib/python3.11/dist-packages/keras/src/backend/tensorflow/nn.py:708: UserWarning: "`sparse\_categorical\_crossentropy` received an invalid argument for dtype: None"
output, from_logits = _get_logits(
6/6 70s 4s/step - accuracy: 0.3108 - loss: 3.7121 - val_accuracy: 0.3571 - val_loss: 142.4979
Epoch 2/10
6/6 18s 3s/step - accuracy: 0.4263 - loss: 2.1424 - val_accuracy: 0.3571 - val_loss: 971.8719
Epoch 3/10
6/6 17s 3s/step - accuracy: 0.7156 - loss: 0.6699 - val_accuracy: 0.3571 - val_loss: 1531.4435
Epoch 4/10
6/6 17s 3s/step - accuracy: 0.8926 - loss: 0.3293 - val_accuracy: 0.3571 - val_loss: 1195.6373
Epoch 5/10
6/6 18s 3s/step - accuracy: 0.9877 - loss: 0.0674 - val_accuracy: 0.3571 - val_loss: 747.2027
Epoch 6/10
6/6 20s 3s/step - accuracy: 1.0000 - loss: 0.0181 - val_accuracy: 0.3571 - val_loss: 343.1469
Epoch 7/10
6/6 18s 3s/step - accuracy: 1.0000 - loss: 0.0075 - val_accuracy: 0.3571 - val_loss: 133.5202
Epoch 8/10
6/6 17s 3s/step - accuracy: 1.0000 - loss: 0.0020 - val_accuracy: 0.3571 - val_loss: 45.3229
Epoch 9/10
6/6 20s 3s/step - accuracy: 1.0000 - loss: 0.0014 - val_accuracy: 0.3571 - val_loss: 13.0383
Epoch 10/10
6/6 18s 3s/step - accuracy: 1.0000 - loss: 5.5965e-04 - val_accuracy: 0.4643 - val_loss: 4.1082

Czas trenowania z GPU wynosi 1 minuta i 20 sekund

```
epochs=10
history = model.fit(
    train_ds,
    validation_data=val_ds,
    epochs=epochs
)

Epoch 1/10
/usr/local/lib/python3.11/dist-packages/keras/src/backend/tensorflow/nn.py:708: UserWarning: "sparse_categorical_crossentropy" output, from_logits = _get_logits(
5/6 ----- 3s 3s/step - accuracy: 0.2170 - loss: 4.3432/usr/local/lib/python3.11/dist-packages/keras/src/backe
output, from_logits = _get_logits(
6/6 ----- 73s 4s/step - accuracy: 0.2315 - loss: 4.3387 - val_accuracy: 0.3571 - val_loss: 39.1711
Epoch 2/10
6/6 ----- 4s 68ms/step - accuracy: 0.4189 - loss: 3.8270 - val_accuracy: 0.3571 - val_loss: 66.6686
Epoch 3/10
6/6 ----- 0s 45ms/step - accuracy: 0.6528 - loss: 1.1346 - val_accuracy: 0.3571 - val_loss: 165.8217
Epoch 4/10
6/6 ----- 0s 45ms/step - accuracy: 0.8520 - loss: 0.3404 - val_accuracy: 0.3571 - val_loss: 130.2823
Epoch 5/10
6/6 ----- 0s 45ms/step - accuracy: 0.9049 - loss: 0.2869 - val_accuracy: 0.3571 - val_loss: 72.6858
Epoch 6/10
6/6 ----- 0s 46ms/step - accuracy: 0.9939 - loss: 0.0212 - val_accuracy: 0.3571 - val_loss: 25.6999
Epoch 7/10
6/6 ----- 0s 45ms/step - accuracy: 0.9912 - loss: 0.0528 - val_accuracy: 0.3571 - val_loss: 8.4135
Epoch 8/10
6/6 ----- 0s 46ms/step - accuracy: 1.0000 - loss: 0.0043 - val_accuracy: 0.3571 - val_loss: 3.0404
Epoch 9/10
6/6 ----- 0s 46ms/step - accuracy: 1.0000 - loss: 0.0016 - val_accuracy: 0.4286 - val_loss: 2.0382
Epoch 10/10
6/6 ----- 0s 46ms/step - accuracy: 1.0000 - loss: 9.1130e-04 - val_accuracy: 0.3929 - val_loss: 1.9037
```

Z porównania wynika, że z GPU czas trenowania modelu jest o wiele krótszy niż czas trenowania bez GPU

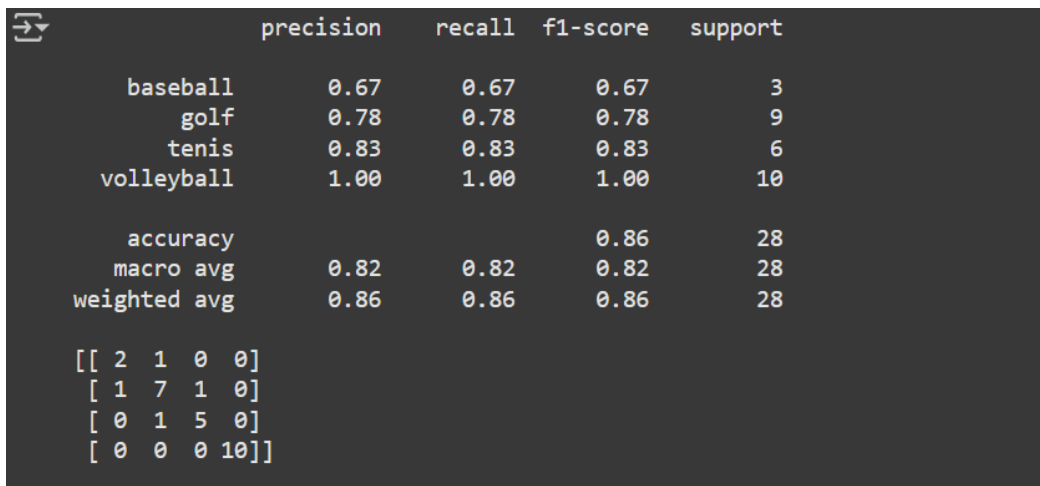
3b. Transfer learning – raport porównawczy bez i z transfer learningiem (badanie poziomu dokładności)

Bez transfer learningu

	precision	recall	f1-score	support
baseball	0.00	0.00	0.00	3
golf	0.50	0.11	0.18	9
tenis	0.40	0.33	0.36	6
volleyball	0.53	1.00	0.69	10
accuracy			0.46	28
macro avg	0.36	0.36	0.31	28
weighted avg	0.43	0.46	0.38	28
[[0 0 1 2]				
[0 1 2 6]				
[2 1 2 1]				
[0 0 0 10]]				

Poziom dokładności wynosi tylko 0,46

Z transfer learningiem



```
precision    recall  f1-score   support

baseball    0.67    0.67    0.67         3
golf        0.78    0.78    0.78         9
tenis       0.83    0.83    0.83         6
volleyball  1.00    1.00    1.00        10

accuracy    0.86    0.86    0.86        28
macro avg   0.82    0.82    0.82        28
weighted avg 0.86    0.86    0.86        28

[[ 2  1  0  0]
 [ 1  7  1  0]
 [ 0  1  5  0]
 [ 0  0  0 10]]
```

Poziom dokładności wynosi 0,86

Poziom dokładności znacząco się poprawił używając transfer learningu.

4a. Normalizacja – raport porównawczy z normalizacją danych i bez

```
batch_size = 20
class_count = 4

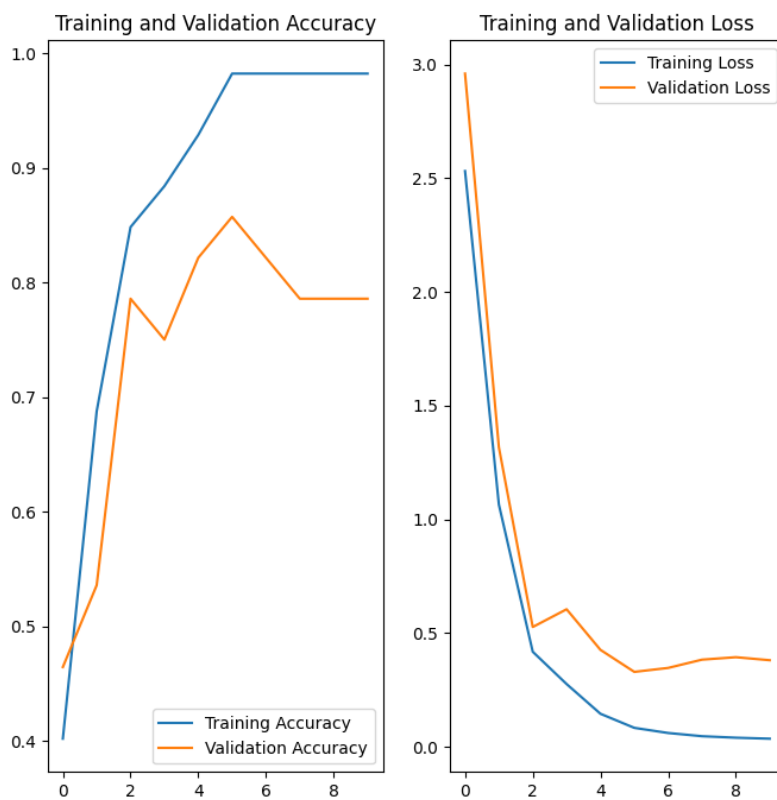
img_height = 64
img_width = 64
```

Używam modelu ResNet50 z transfer learningiem (GPU)

	Czas treningu	Poziom dokładności
BEZ NORMALIZACJI	20 sek.	0,79
Z NORMALIZACJĄ	23 sek.	0,89

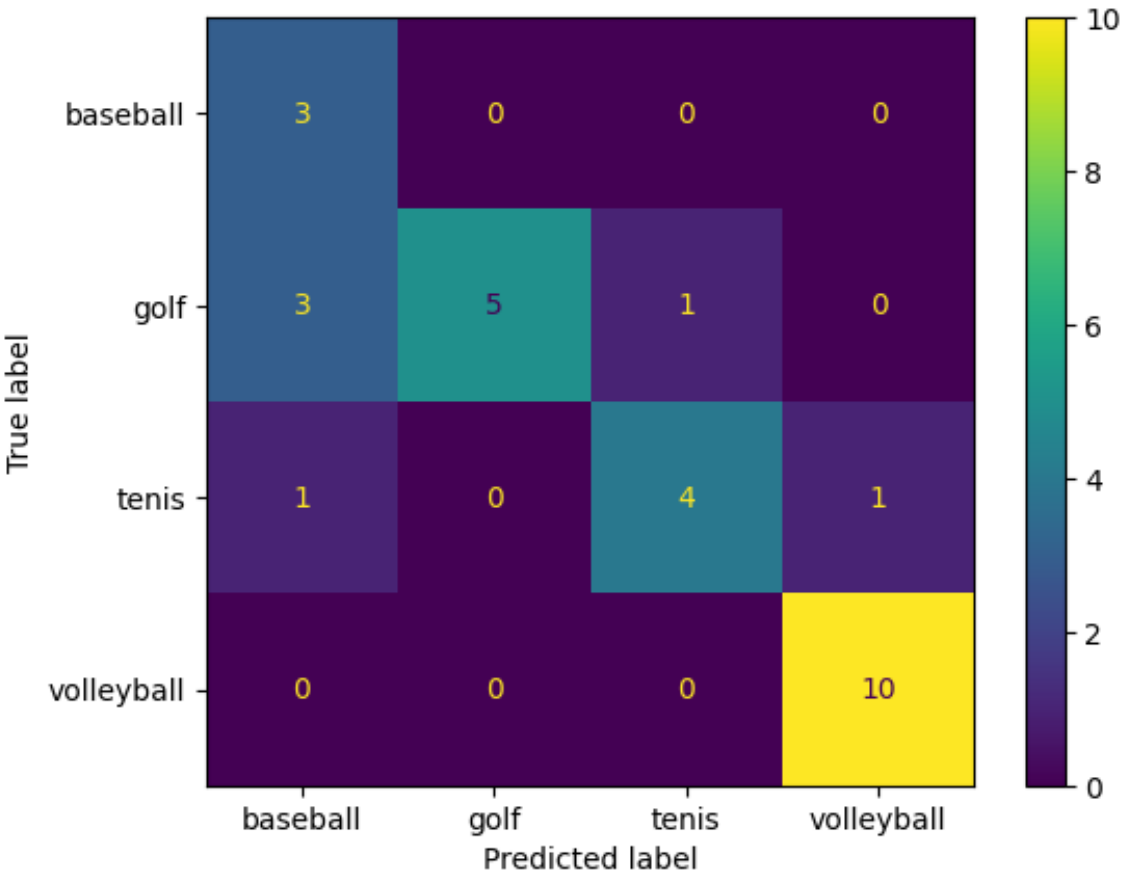
Bez normalizacji:

```
Epoch 1/10
/usr/local/lib/python3.11/dist-packages/keras/src/backend/tensorflow/nn.py:708: UserWarning: "sparse_categorical_crossentropy" is deprecated. Use "sparse_categorical_crossentropy" instead.
  output, from_logits = _get_logits(
6/6 ----- 19s 2s/step - accuracy: 0.3438 - loss: 3.2802 - val_accuracy: 0.4643 - val_loss: 2.9602
Epoch 2/10
6/6 ----- 0s 34ms/step - accuracy: 0.6145 - loss: 1.4576 - val_accuracy: 0.5357 - val_loss: 1.3168
Epoch 3/10
6/6 ----- 0s 22ms/step - accuracy: 0.8469 - loss: 0.5101 - val_accuracy: 0.7857 - val_loss: 0.5271
Epoch 4/10
6/6 ----- 0s 28ms/step - accuracy: 0.8831 - loss: 0.3281 - val_accuracy: 0.7500 - val_loss: 0.6052
Epoch 5/10
6/6 ----- 0s 18ms/step - accuracy: 0.9223 - loss: 0.1741 - val_accuracy: 0.8214 - val_loss: 0.4267
Epoch 6/10
6/6 ----- 0s 18ms/step - accuracy: 0.9841 - loss: 0.0777 - val_accuracy: 0.8571 - val_loss: 0.3298
Epoch 7/10
6/6 ----- 0s 15ms/step - accuracy: 0.9835 - loss: 0.0667 - val_accuracy: 0.8214 - val_loss: 0.3472
Epoch 8/10
6/6 ----- 0s 15ms/step - accuracy: 0.9933 - loss: 0.0353 - val_accuracy: 0.7857 - val_loss: 0.3836
Epoch 9/10
6/6 ----- 0s 14ms/step - accuracy: 0.9933 - loss: 0.0270 - val_accuracy: 0.7857 - val_loss: 0.3944
Epoch 10/10
6/6 ----- 0s 14ms/step - accuracy: 0.9783 - loss: 0.0451 - val_accuracy: 0.7857 - val_loss: 0.3811
```



	precision	recall	f1-score	support
baseball	0.43	1.00	0.60	3
golf	1.00	0.56	0.71	9
tenis	0.80	0.67	0.73	6
volleyball	0.91	1.00	0.95	10
accuracy			0.79	28
macro avg	0.78	0.81	0.75	28
weighted avg	0.86	0.79	0.79	28

[[3 0 0 0]
[3 5 1 0]
[1 0 4 1]
[0 0 0 10]]



Z normalizacją:

```
[14] epochs=10
      history = model.fit(
        train_ds,
        validation_data=val_ds,
        epochs=epochs
      )
```

Epoch 1/10
/usr/local/lib/python3.11/dist-packages/keras/src/backend/tensorflow/nn.py:708: UserWarning: "`sparse_categorical_crossentropy, from_logits = _get_logits(
6/6 22s 2s/step - accuracy: 0.2263 - loss: 4.2603 - val_accuracy: 0.5357 - val_loss: 3.3367

Epoch 2/10
6/6 0s 28ms/step - accuracy: 0.7348 - loss: 0.7894 - val_accuracy: 0.8571 - val_loss: 0.8967

Epoch 3/10
6/6 0s 33ms/step - accuracy: 0.9603 - loss: 0.1543 - val_accuracy: 0.8214 - val_loss: 0.6812

Epoch 4/10
6/6 0s 24ms/step - accuracy: 0.9443 - loss: 0.0953 - val_accuracy: 0.8571 - val_loss: 1.0505

Epoch 5/10
6/6 0s 30ms/step - accuracy: 1.0000 - loss: 0.0202 - val_accuracy: 0.8214 - val_loss: 1.0896

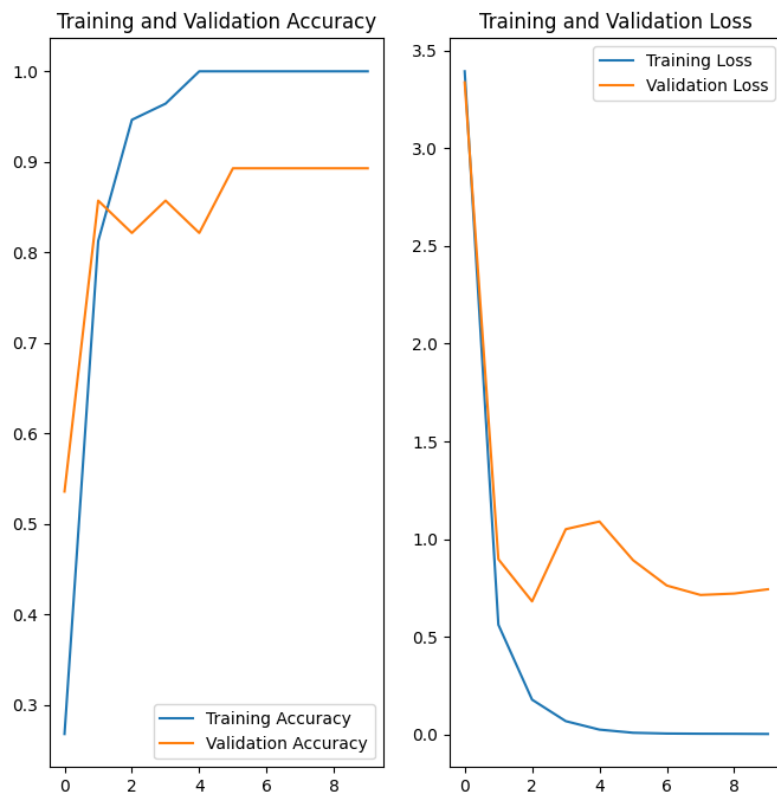
Epoch 6/10
6/6 0s 24ms/step - accuracy: 1.0000 - loss: 0.0089 - val_accuracy: 0.8929 - val_loss: 0.8915

Epoch 7/10
6/6 0s 24ms/step - accuracy: 1.0000 - loss: 0.0039 - val_accuracy: 0.8929 - val_loss: 0.7620

Epoch 8/10
6/6 0s 22ms/step - accuracy: 1.0000 - loss: 0.0025 - val_accuracy: 0.8929 - val_loss: 0.7138

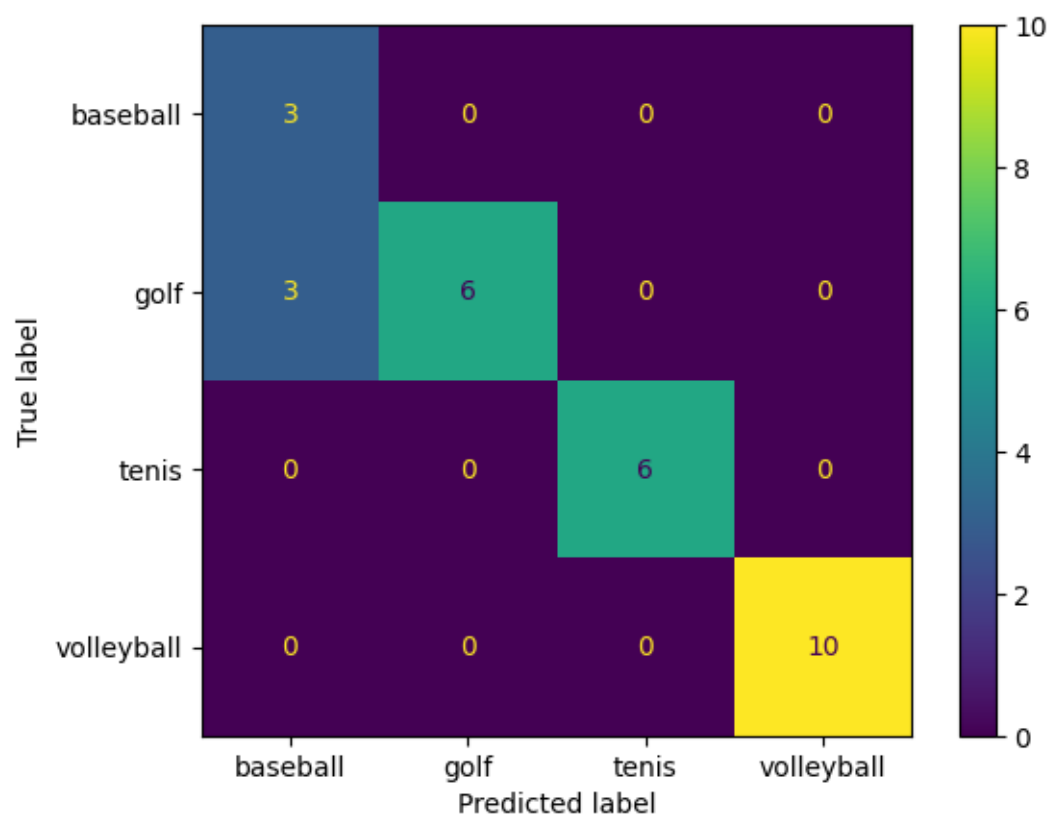
Epoch 9/10
6/6 0s 27ms/step - accuracy: 1.0000 - loss: 0.0050 - val_accuracy: 0.8929 - val_loss: 0.7210

Epoch 10/10
6/6 0s 17ms/step - accuracy: 1.0000 - loss: 0.0025 - val_accuracy: 0.8929 - val_loss: 0.7430



	precision	recall	f1-score	support
baseball	0.50	1.00	0.67	3
golf	1.00	0.67	0.80	9
tenis	1.00	1.00	1.00	6
volleyball	1.00	1.00	1.00	10
accuracy			0.89	28
macro avg	0.88	0.92	0.87	28
weighted avg	0.95	0.89	0.90	28

[[3 0 0 0]
[3 6 0 0]
[0 0 6 0]
[0 0 0 10]]



4b. Augmentacja danych – raport porównawczy z augmentacją i bez

```
batch_size = 20
class_count = 4

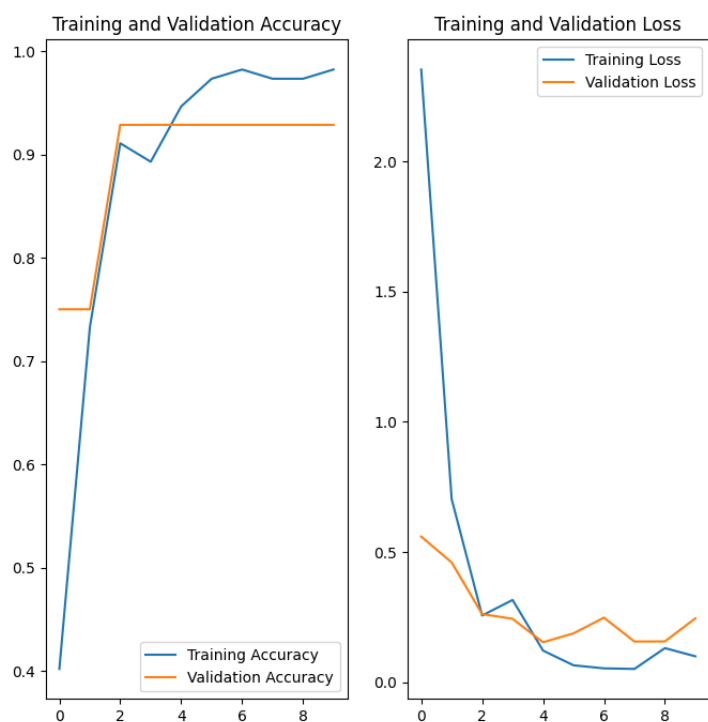
img_height = 64
img_width = 64
```

Używam modelu ResNet50 z transfer learningiem (GPU)

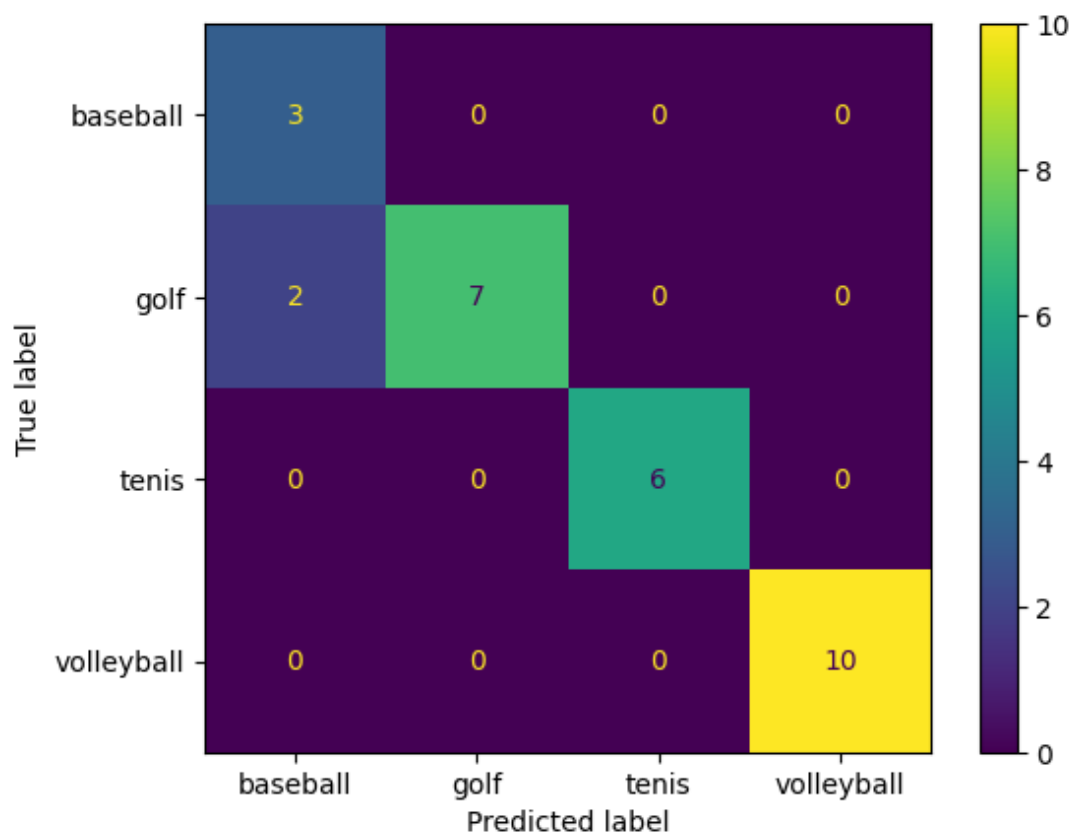
	Czas treningu	Poziom dokładności
BEZ AUGMENTACJI	20 sek.	0,79
Z AUGMENTACJĄ	39 sek.	0,93

Z augmentacją wyniki:

```
Epoch 1/10
/usr/local/lib/python3.11/dist-packages/keras/src/backend/tensorflow/nn.py:708: UserWarning: ``sparse_categorical_crossentropy``
output, from_logits = _get_logits(
6/6 ██████████ 16s 967ms/step - accuracy: 0.3333 - loss: 2.7119 - val_accuracy: 0.7500 - val_loss: 0.5591
Epoch 2/10
6/6 ██████████ 2s 316ms/step - accuracy: 0.6938 - loss: 0.7929 - val_accuracy: 0.7500 - val_loss: 0.4597
Epoch 3/10
6/6 ██████████ 2s 339ms/step - accuracy: 0.9280 - loss: 0.1855 - val_accuracy: 0.9286 - val_loss: 0.2612
Epoch 4/10
6/6 ██████████ 3s 457ms/step - accuracy: 0.9221 - loss: 0.2454 - val_accuracy: 0.9286 - val_loss: 0.2430
Epoch 5/10
6/6 ██████████ 4s 320ms/step - accuracy: 0.9398 - loss: 0.1310 - val_accuracy: 0.9286 - val_loss: 0.1530
Epoch 6/10
6/6 ██████████ 2s 321ms/step - accuracy: 0.9532 - loss: 0.0936 - val_accuracy: 0.9286 - val_loss: 0.1870
Epoch 7/10
6/6 ██████████ 2s 340ms/step - accuracy: 0.9914 - loss: 0.0335 - val_accuracy: 0.9286 - val_loss: 0.2474
Epoch 8/10
6/6 ██████████ 2s 408ms/step - accuracy: 0.9726 - loss: 0.0533 - val_accuracy: 0.9286 - val_loss: 0.1558
Epoch 9/10
6/6 ██████████ 3s 564ms/step - accuracy: 0.9877 - loss: 0.0663 - val_accuracy: 0.9286 - val_loss: 0.1562
Epoch 10/10
6/6 ██████████ 2s 317ms/step - accuracy: 0.9949 - loss: 0.0424 - val_accuracy: 0.9286 - val_loss: 0.2448
```



	precision	recall	f1-score	support
baseball	0.60	1.00	0.75	3
golf	1.00	0.78	0.88	9
tenis	1.00	1.00	1.00	6
volleyball	1.00	1.00	1.00	10
accuracy			0.93	28
macro avg	0.90	0.94	0.91	28
weighted avg	0.96	0.93	0.93	28
[[3 0 0 0]				
[2 7 0 0]				
[0 0 6 0]				
[0 0 0 10]]				

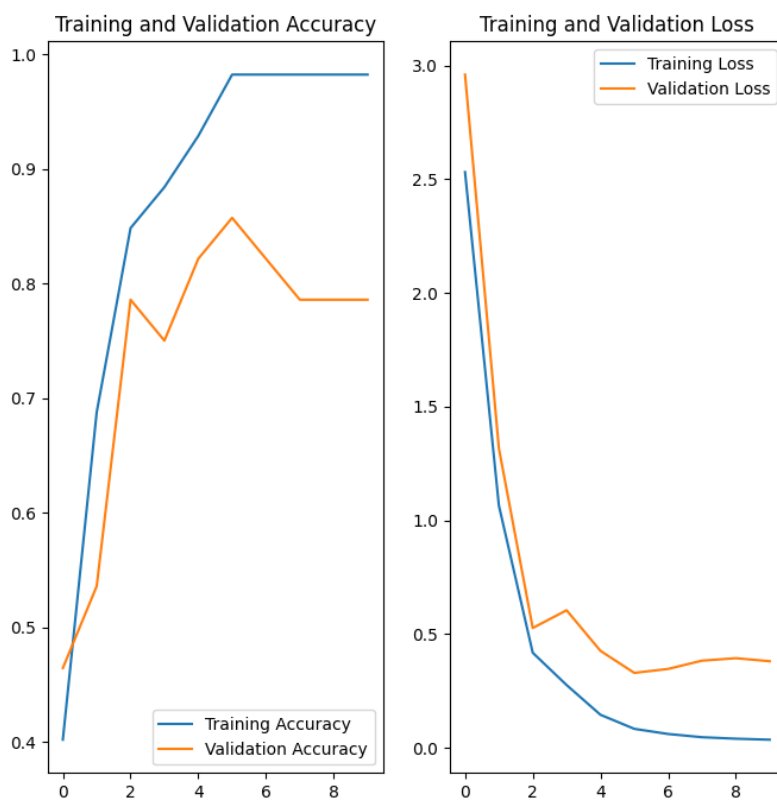


Przeprowadzone przekształcenia

Augmentacja	Opis	Cel
RandomFlip("horizontal")	Odbicie poziome	Rozpoznawanie w różnych kierunkach
RandomRotation(0.1)	Obrót $\pm 36^\circ$	Tolerancja na przechylenie obrazu
RandomZoom(0.1)	Zoom $\pm 10\%$	Różne rozmiary obiektów
RandomTranslation(0.1, 0.1)	Przesunięcie $\pm 10\%$	Różne pozycje obiektu na obrazie
RandomContrast(0.2)	Zmiana kontrastu $\pm 20\%$	Oświetlenie, jakość zdjęcia
RandomBrightness(0.2)	Zmiana jasności $\pm 20\%$	Warunki oświetleniowe (światło/cień)

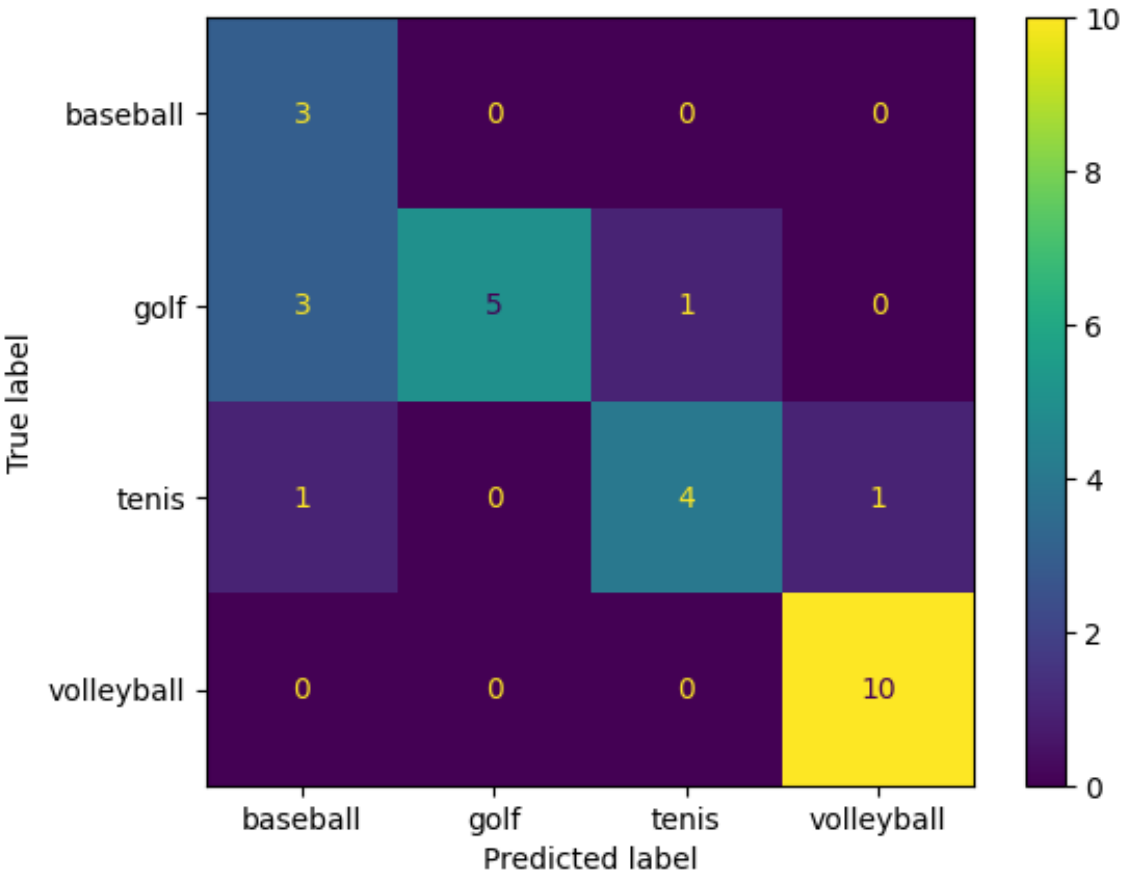
Bez augmentacji wyniki:

```
Epoch 1/10  
/usr/local/lib/python3.11/dist-packages/keras/src/backend/tensorflow/nn.py:708: UserWarning: "sparse_categorical_crossentropy" is deprecated. Use "sparse_categorical_crossentropy" instead.  
output, from_logits = _get_logits(  
6/6 ----- 19s 2s/step - accuracy: 0.3438 - loss: 3.2802 - val_accuracy: 0.4643 - val_loss: 2.9602  
Epoch 2/10  
6/6 ----- 0s 34ms/step - accuracy: 0.6145 - loss: 1.4576 - val_accuracy: 0.5357 - val_loss: 1.3168  
Epoch 3/10  
6/6 ----- 0s 22ms/step - accuracy: 0.8469 - loss: 0.5101 - val_accuracy: 0.7857 - val_loss: 0.5271  
Epoch 4/10  
6/6 ----- 0s 28ms/step - accuracy: 0.8831 - loss: 0.3281 - val_accuracy: 0.7500 - val_loss: 0.6052  
Epoch 5/10  
6/6 ----- 0s 18ms/step - accuracy: 0.9223 - loss: 0.1741 - val_accuracy: 0.8214 - val_loss: 0.4267  
Epoch 6/10  
6/6 ----- 0s 18ms/step - accuracy: 0.9841 - loss: 0.0777 - val_accuracy: 0.8571 - val_loss: 0.3298  
Epoch 7/10  
6/6 ----- 0s 15ms/step - accuracy: 0.9835 - loss: 0.0667 - val_accuracy: 0.8214 - val_loss: 0.3472  
Epoch 8/10  
6/6 ----- 0s 15ms/step - accuracy: 0.9933 - loss: 0.0353 - val_accuracy: 0.7857 - val_loss: 0.3836  
Epoch 9/10  
6/6 ----- 0s 14ms/step - accuracy: 0.9933 - loss: 0.0270 - val_accuracy: 0.7857 - val_loss: 0.3944  
Epoch 10/10  
6/6 ----- 0s 14ms/step - accuracy: 0.9783 - loss: 0.0451 - val_accuracy: 0.7857 - val_loss: 0.3811
```



	precision	recall	f1-score	support
baseball	0.43	1.00	0.60	3
golf	1.00	0.56	0.71	9
tenis	0.80	0.67	0.73	6
volleyball	0.91	1.00	0.95	10
accuracy			0.79	28
macro avg	0.78	0.81	0.75	28
weighted avg	0.86	0.79	0.79	28

[[	3	0	0	0]
[	3	5	1	0]
[	1	0	4	1]
[	0	0	0	10]]



4c. Dropout – raport porównawczy z zastosowaniem dropoutu i bez

Używam modelu ResNet50 z transfer learningiem (GPU)

	CZAS	DOKŁADNOŚĆ
Dropout	40 sek.	0,82
Bez dropoutu	20 sek.	0,79

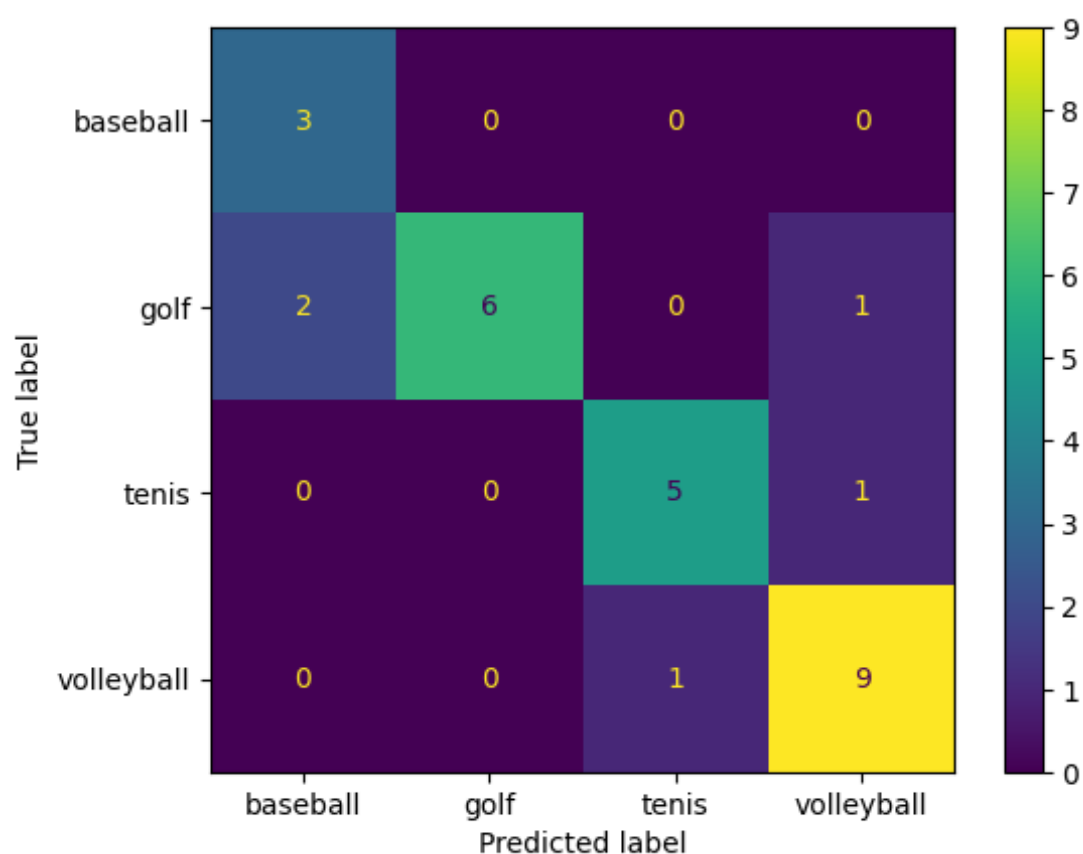
Dropout:

```
Epoch 1/10
/usr/local/lib/python3.11/dist-packages/keras/src/backend/tensorflow/nn.py:708: UserWarning: "sparse_categorical_crossentropy" output, from_logits = _get_logits(
6/6 16s 960ms/step - accuracy: 0.2592 - loss: 3.7546 - val_accuracy: 0.5000 - val_loss: 2.0423
Epoch 2/10
6/6 3s 549ms/step - accuracy: 0.5134 - loss: 1.8809 - val_accuracy: 0.7143 - val_loss: 1.3994
Epoch 3/10
6/6 4s 336ms/step - accuracy: 0.6600 - loss: 1.2828 - val_accuracy: 0.6786 - val_loss: 1.3245
Epoch 4/10
6/6 3s 337ms/step - accuracy: 0.7932 - loss: 0.8701 - val_accuracy: 0.8214 - val_loss: 0.7786
Epoch 5/10
6/6 2s 333ms/step - accuracy: 0.8074 - loss: 0.7649 - val_accuracy: 0.7857 - val_loss: 0.6721
Epoch 6/10
6/6 4s 553ms/step - accuracy: 0.8807 - loss: 0.2990 - val_accuracy: 0.7500 - val_loss: 0.8661
Epoch 7/10
6/6 4s 327ms/step - accuracy: 0.9147 - loss: 0.2813 - val_accuracy: 0.7500 - val_loss: 0.8924
Epoch 8/10
6/6 2s 331ms/step - accuracy: 0.8382 - loss: 0.3589 - val_accuracy: 0.7857 - val_loss: 0.7164
Epoch 9/10
6/6 2s 331ms/step - accuracy: 0.8942 - loss: 0.2930 - val_accuracy: 0.7857 - val_loss: 0.5718
Epoch 10/10
6/6 2s 378ms/step - accuracy: 0.9561 - loss: 0.1083 - val_accuracy: 0.8214 - val_loss: 0.4483
```



	precision	recall	f1-score	support
baseball	0.60	1.00	0.75	3
golf	1.00	0.67	0.80	9
tenis	0.83	0.83	0.83	6
volleyball	0.82	0.90	0.86	10
accuracy			0.82	28
macro avg	0.81	0.85	0.81	28
weighted avg	0.86	0.82	0.82	28


```
[[3 0 0 0]
 [2 6 0 1]
 [0 0 5 1]
 [0 0 1 9]]
```



4d. Dokładanie danych – raport porównawczy z dołożeniem nowej partii danych i bez

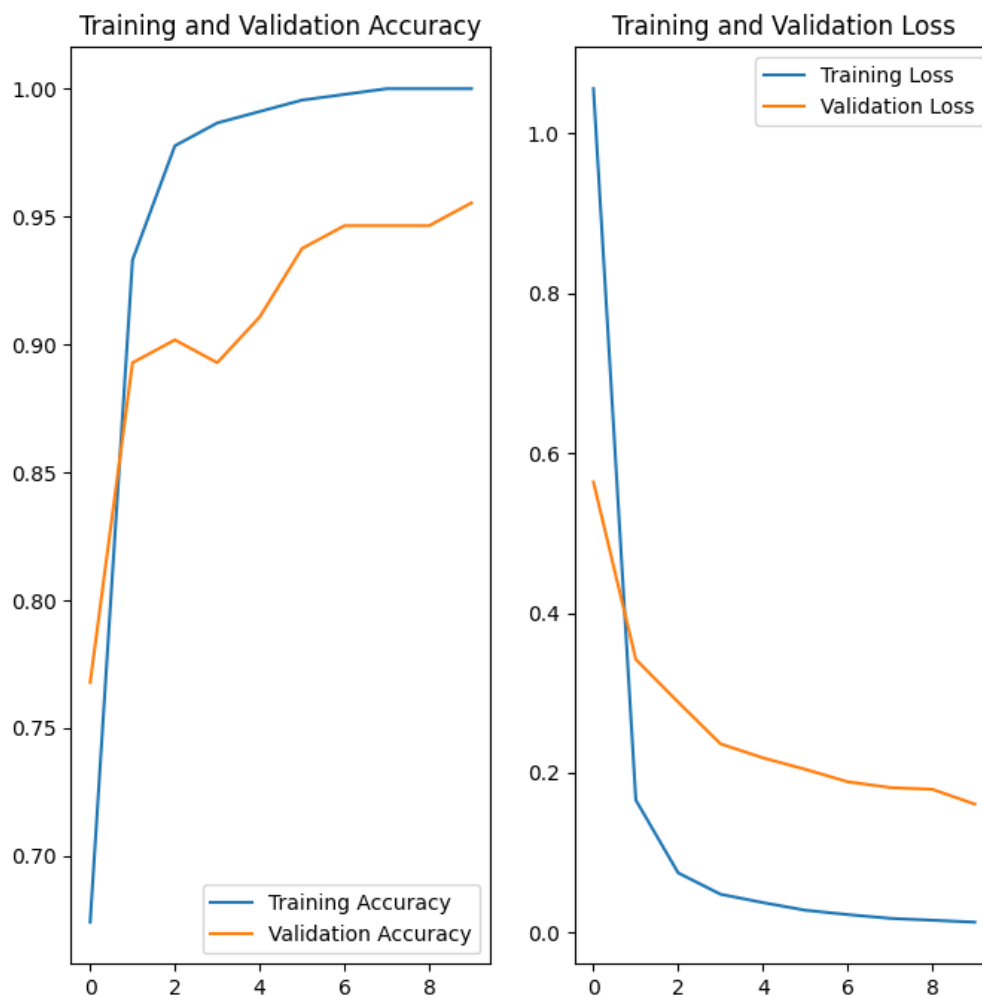
```
batch_size = 20  
class_count = 4  
  
img_height = 64  
img_width = 64
```

	CZAS	DOKŁADNOŚĆ
Z nową partią danych	2 min. 14 sek.	0,96
Bez nowej partii danych	20 sek.	0,79

Z nową partią danych:

```
[46] epochs=10
     history = model.fit(
         train_ds,
         validation_data=val_ds,
         epochs=epochs
     )
```

Epoch 1/10
23/23 37s 699ms/step - accuracy: 0.6005 - loss: 1.4779 - val_accuracy: 0.7679 - val_loss: 0.5639
Epoch 2/10
23/23 23s 391ms/step - accuracy: 0.9047 - loss: 0.2005 - val_accuracy: 0.8929 - val_loss: 0.3419
Epoch 3/10
23/23 9s 388ms/step - accuracy: 0.9802 - loss: 0.0624 - val_accuracy: 0.9018 - val_loss: 0.2884
Epoch 4/10
23/23 10s 382ms/step - accuracy: 0.9929 - loss: 0.0355 - val_accuracy: 0.8929 - val_loss: 0.2362
Epoch 5/10
23/23 10s 385ms/step - accuracy: 0.9892 - loss: 0.0426 - val_accuracy: 0.9107 - val_loss: 0.2187
Epoch 6/10
23/23 9s 382ms/step - accuracy: 0.9984 - loss: 0.0231 - val_accuracy: 0.9375 - val_loss: 0.2043
Epoch 7/10
23/23 9s 377ms/step - accuracy: 0.9959 - loss: 0.0253 - val_accuracy: 0.9464 - val_loss: 0.1888
Epoch 8/10
23/23 9s 346ms/step - accuracy: 1.0000 - loss: 0.0199 - val_accuracy: 0.9464 - val_loss: 0.1815
Epoch 9/10
23/23 9s 408ms/step - accuracy: 1.0000 - loss: 0.0192 - val_accuracy: 0.9464 - val_loss: 0.1793
Epoch 10/10
23/23 9s 377ms/step - accuracy: 1.0000 - loss: 0.0144 - val_accuracy: 0.9554 - val_loss: 0.1608

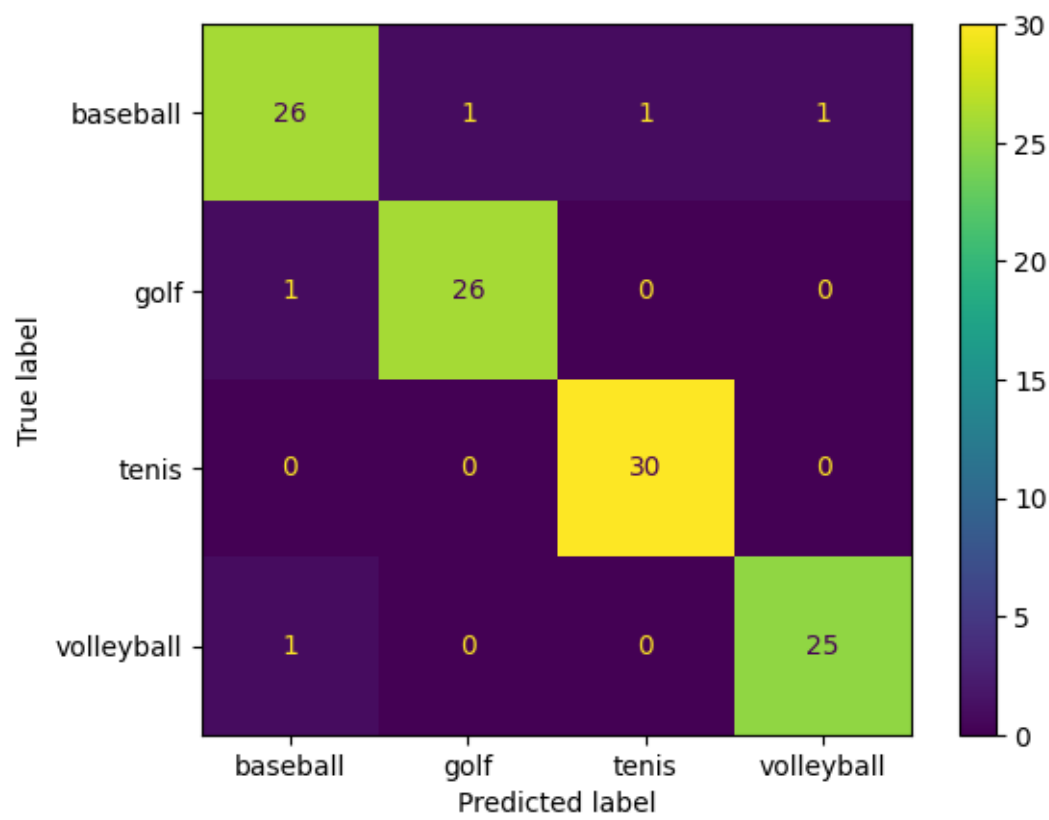


	precision	recall	f1-score	support
baseball	0.93	0.90	0.91	29
golf	0.96	0.96	0.96	27
tenis	0.97	1.00	0.98	30
volleyball	0.96	0.96	0.96	26
accuracy			0.96	112
macro avg	0.96	0.96	0.96	112
weighted avg	0.96	0.96	0.96	112


```

[[26  1  1  1]
 [ 1 26  0  0]
 [ 0  0 30  0]
 [ 1  0  0 25]]

```



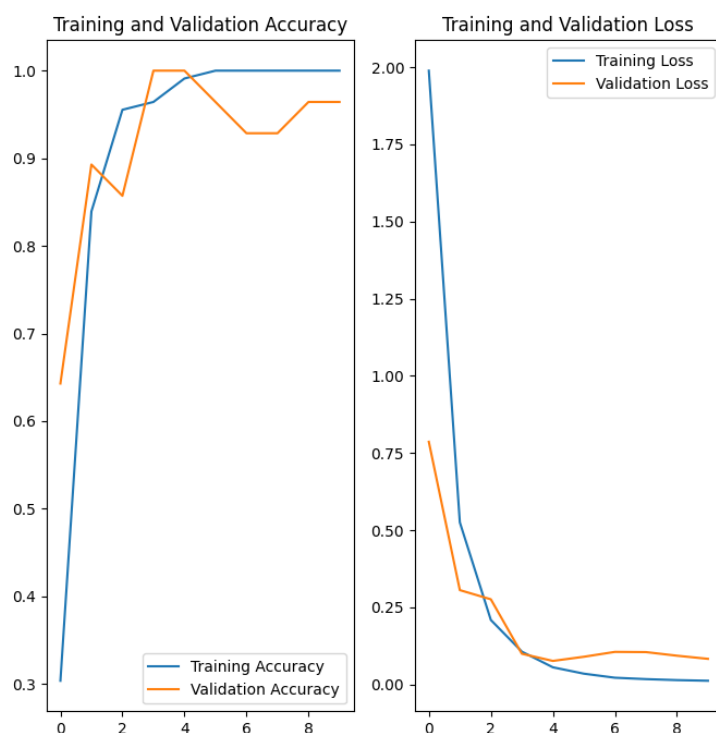
4e. Różne rozmiary wejściowe (np. 96x96, 160x160, 224x224) – raport porównawczy dla każdego rozmiaru

	CZAS	DOKŁADNOŚĆ
96x96	20 sek.	0,96
160x160	22 sek.	1,00
224x224	33 sek.	1,00

96x96:

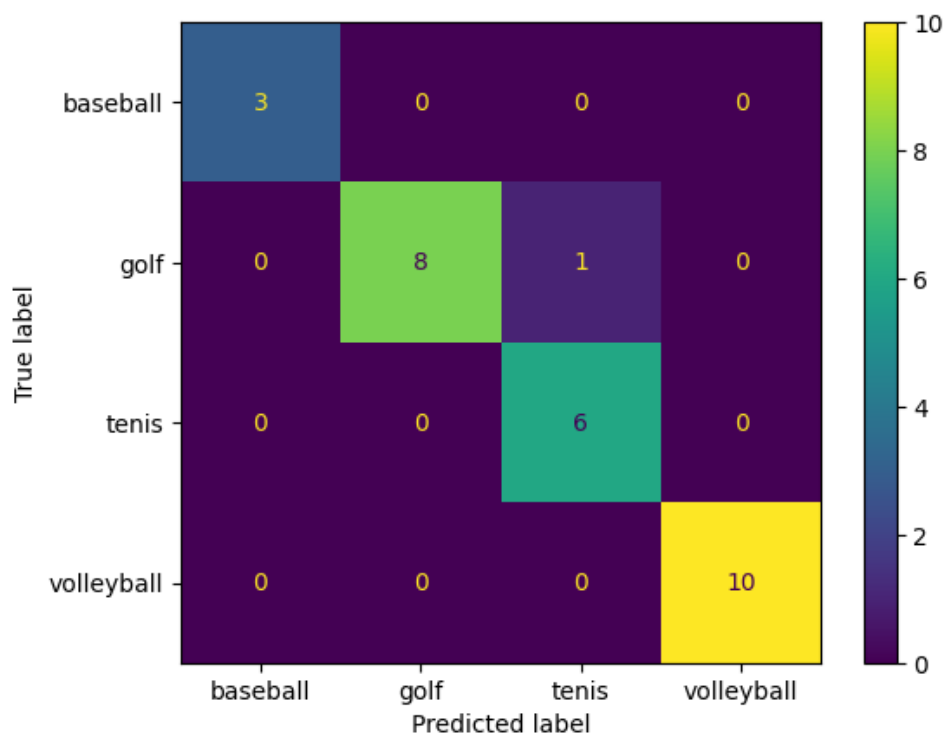
```
[53] epochs=10
history = model.fit(
    train_ds,
    validation_data=val_ds,
    epochs=epochs
)

Epoch 1/10
/usr/local/lib/python3.11/dist-packages/keras/src/backend/tensorflow/nn.py:708: UserWarning: "sparse_categorical_crossentropy" output, from_logits = _get_logits(
6/6 18s 2s/step - accuracy: 0.2238 - loss: 2.3603 - val_accuracy: 0.6429 - val_loss: 0.7863
Epoch 2/10
6/6 0s 33ms/step - accuracy: 0.8412 - loss: 0.6542 - val_accuracy: 0.8929 - val_loss: 0.3057
Epoch 3/10
6/6 0s 33ms/step - accuracy: 0.9569 - loss: 0.2581 - val_accuracy: 0.8571 - val_loss: 0.2760
Epoch 4/10
6/6 0s 24ms/step - accuracy: 0.9501 - loss: 0.1284 - val_accuracy: 1.0000 - val_loss: 0.0995
Epoch 5/10
6/6 0s 23ms/step - accuracy: 0.9939 - loss: 0.0444 - val_accuracy: 1.0000 - val_loss: 0.0763
Epoch 6/10
6/6 0s 24ms/step - accuracy: 1.0000 - loss: 0.0458 - val_accuracy: 0.9643 - val_loss: 0.0898
Epoch 7/10
6/6 0s 32ms/step - accuracy: 1.0000 - loss: 0.0283 - val_accuracy: 0.9286 - val_loss: 0.1056
Epoch 8/10
6/6 0s 31ms/step - accuracy: 1.0000 - loss: 0.0201 - val_accuracy: 0.9286 - val_loss: 0.1049
Epoch 9/10
6/6 0s 25ms/step - accuracy: 1.0000 - loss: 0.0134 - val_accuracy: 0.9643 - val_loss: 0.0932
Epoch 10/10
6/6 0s 31ms/step - accuracy: 1.0000 - loss: 0.0122 - val_accuracy: 0.9643 - val_loss: 0.0830
```



	precision	recall	f1-score	support
baseball	1.00	1.00	1.00	3
golf	1.00	0.89	0.94	9
tenis	0.86	1.00	0.92	6
volleyball	1.00	1.00	1.00	10
accuracy			0.96	28
macro avg	0.96	0.97	0.97	28
weighted avg	0.97	0.96	0.96	28


```
[[ 3  0  0  0]
 [ 0  8  1  0]
 [ 0  0  6  0]
 [ 0  0  0 10]]
```

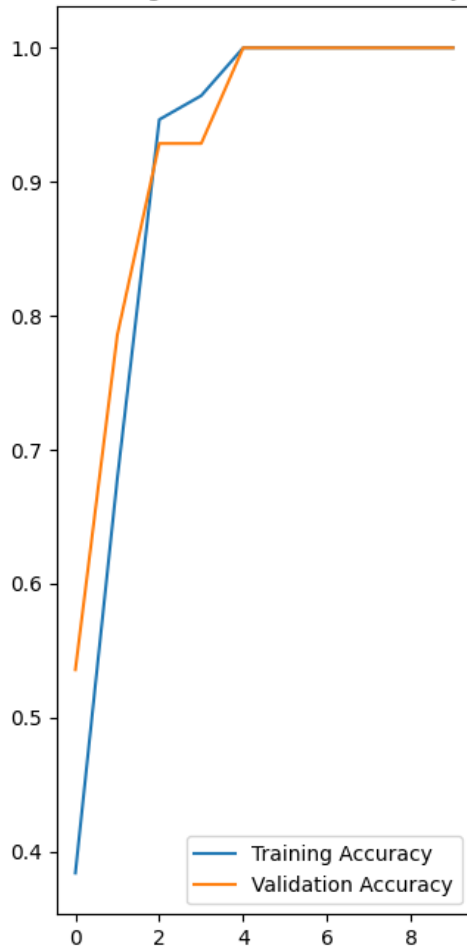


160x160:

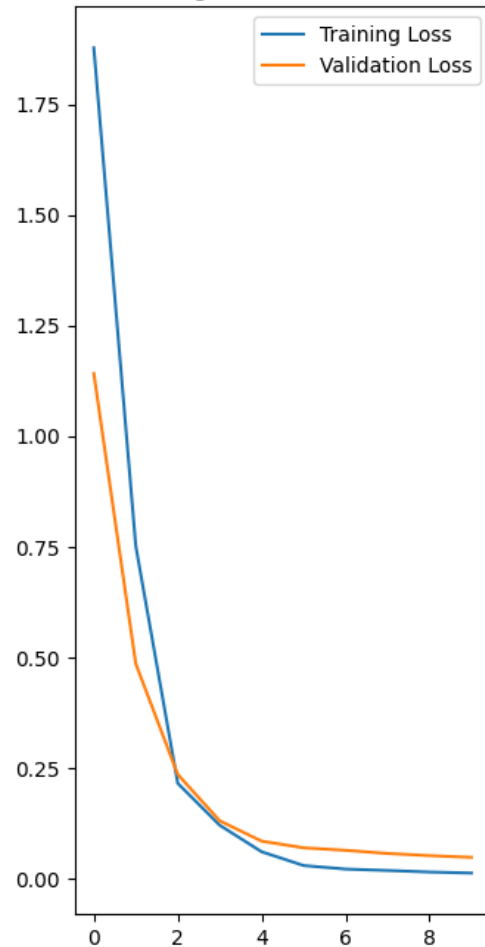
```
[33] epochs=10
history = model.fit(
    train_ds,
    validation_data=val_ds,
    epochs=epochs
)
```

Epoch 1/10
/usr/local/lib/python3.11/dist-packages/keras/src/backend/tensorflow/nn.py:708: UserWarning: "`sparse_categorical_crossentropy, from_logits = _get_logits(
6/6 ----- 20s 2s/step - accuracy: 0.3285 - loss: 2.1322 - val_accuracy: 0.5357 - val_loss: 1.1418
Epoch 2/10
6/6 ----- 0s 52ms/step - accuracy: 0.6496 - loss: 0.8097 - val_accuracy: 0.7857 - val_loss: 0.4860
Epoch 3/10
6/6 ----- 0s 42ms/step - accuracy: 0.9124 - loss: 0.2671 - val_accuracy: 0.9286 - val_loss: 0.2358
Epoch 4/10
6/6 ----- 0s 41ms/step - accuracy: 0.9655 - loss: 0.1266 - val_accuracy: 0.9286 - val_loss: 0.1310
Epoch 5/10
6/6 ----- 0s 42ms/step - accuracy: 1.0000 - loss: 0.0791 - val_accuracy: 1.0000 - val_loss: 0.0850
Epoch 6/10
6/6 ----- 0s 45ms/step - accuracy: 1.0000 - loss: 0.0316 - val_accuracy: 1.0000 - val_loss: 0.0700
Epoch 7/10
6/6 ----- 0s 47ms/step - accuracy: 1.0000 - loss: 0.0198 - val_accuracy: 1.0000 - val_loss: 0.0643
Epoch 8/10
6/6 ----- 0s 42ms/step - accuracy: 1.0000 - loss: 0.0235 - val_accuracy: 1.0000 - val_loss: 0.0572
Epoch 9/10
6/6 ----- 0s 43ms/step - accuracy: 1.0000 - loss: 0.0142 - val_accuracy: 1.0000 - val_loss: 0.0523
Epoch 10/10
6/6 ----- 0s 41ms/step - accuracy: 1.0000 - loss: 0.0120 - val_accuracy: 1.0000 - val_loss: 0.0484

Training and Validation Accuracy

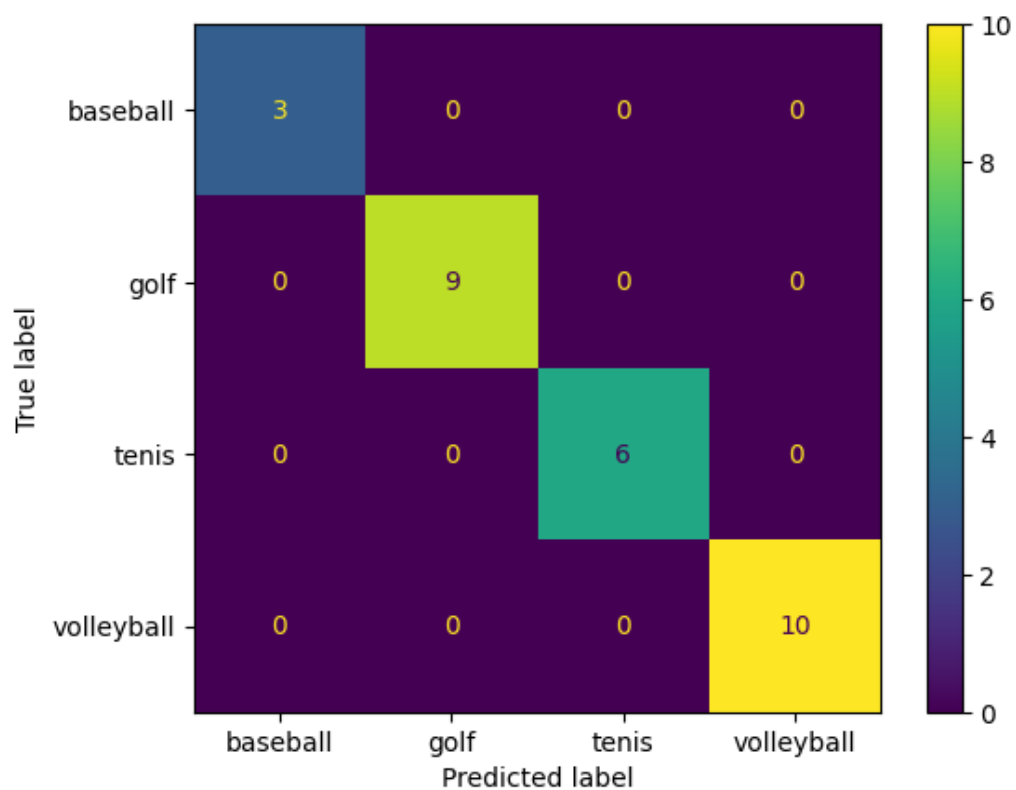


Training and Validation Loss



	precision	recall	f1-score	support
baseball	1.00	1.00	1.00	3
golf	1.00	1.00	1.00	9
tenis	1.00	1.00	1.00	6
volleyball	1.00	1.00	1.00	10
accuracy			1.00	28
macro avg	1.00	1.00	1.00	28
weighted avg	1.00	1.00	1.00	28

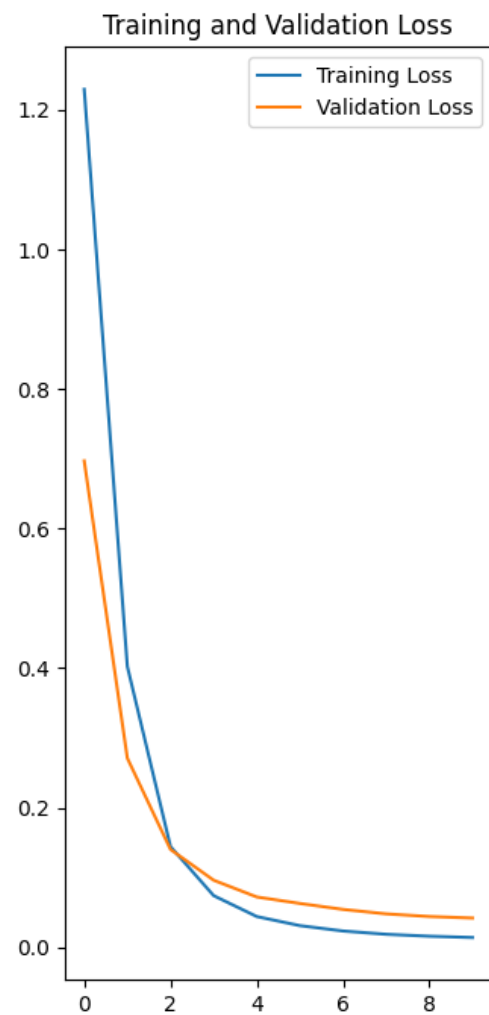
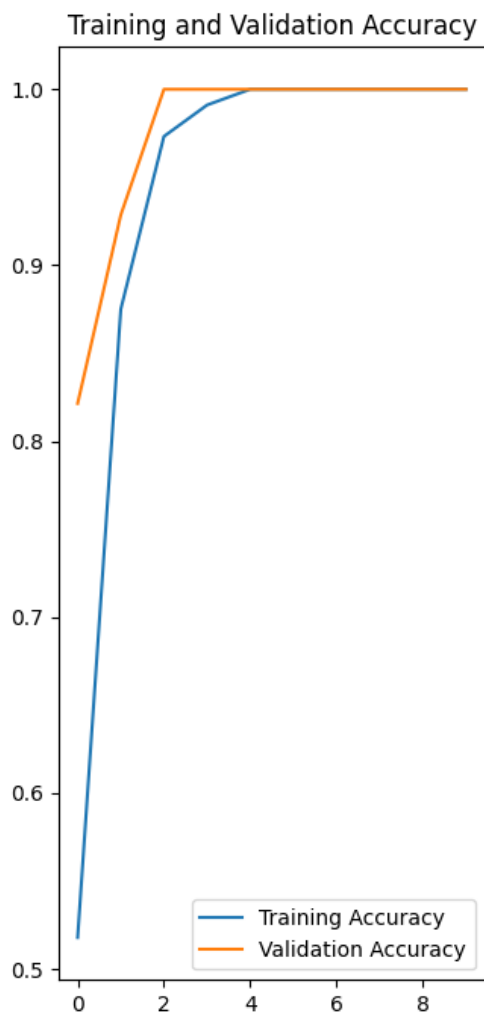
[[3 0 0 0]
[0 9 0 0]
[0 0 6 0]
[0 0 0 10]]



224x224:

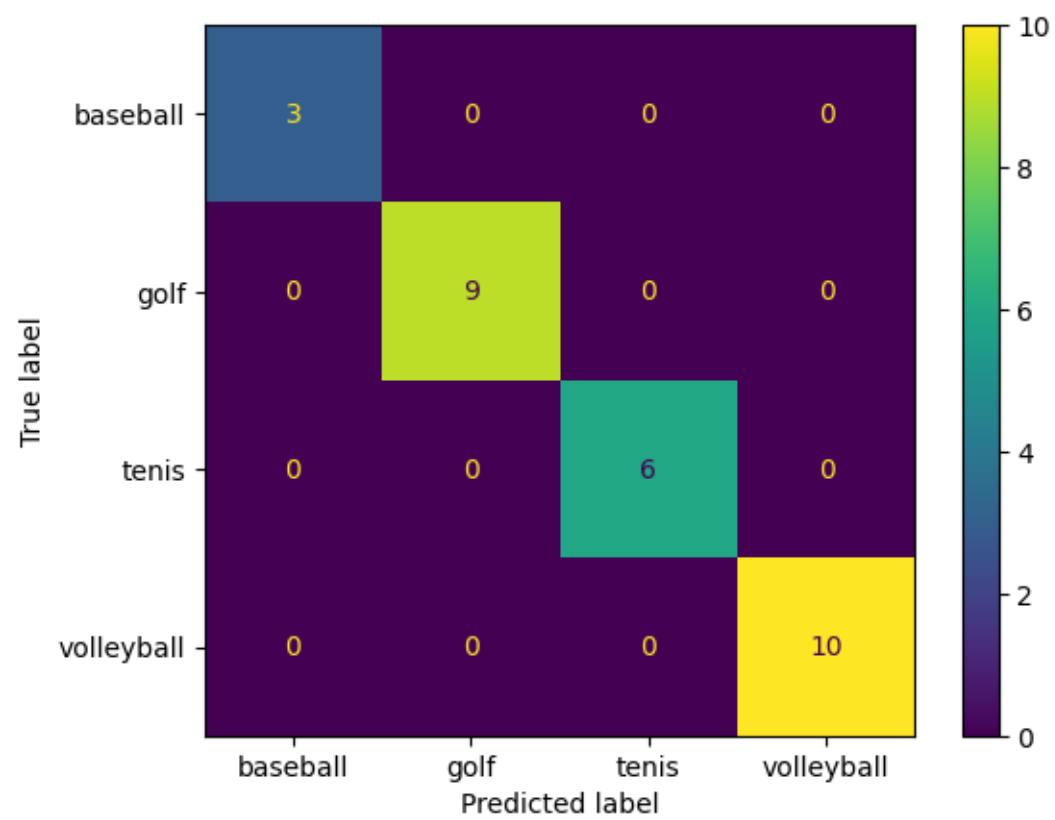
```
[13] epochs=10
     history = model.fit(
         train_ds,
         validation_data=val_ds,
         epochs=epochs
     )

Epoch 1/10
/usr/local/lib/python3.11/dist-packages/keras/src/backend/tensorflow/nn.py:708: UserWarning: "sparse_categorical_crossentropy" output, from_logits = _get_logits(
6/6 ----- 31s 2s/step - accuracy: 0.4460 - loss: 1.4095 - val_accuracy: 0.8214 - val_loss: 0.6969
Epoch 2/10
6/6 ----- 0s 64ms/step - accuracy: 0.8227 - loss: 0.4933 - val_accuracy: 0.9286 - val_loss: 0.2710
Epoch 3/10
6/6 ----- 0s 64ms/step - accuracy: 0.9821 - loss: 0.1662 - val_accuracy: 1.0000 - val_loss: 0.1402
Epoch 4/10
6/6 ----- 0s 63ms/step - accuracy: 0.9880 - loss: 0.0841 - val_accuracy: 1.0000 - val_loss: 0.0962
Epoch 5/10
6/6 ----- 0s 66ms/step - accuracy: 1.0000 - loss: 0.0467 - val_accuracy: 1.0000 - val_loss: 0.0721
Epoch 6/10
6/6 ----- 1s 65ms/step - accuracy: 1.0000 - loss: 0.0373 - val_accuracy: 1.0000 - val_loss: 0.0628
Epoch 7/10
6/6 ----- 1s 79ms/step - accuracy: 1.0000 - loss: 0.0278 - val_accuracy: 1.0000 - val_loss: 0.0543
Epoch 8/10
6/6 ----- 0s 64ms/step - accuracy: 1.0000 - loss: 0.0208 - val_accuracy: 1.0000 - val_loss: 0.0479
Epoch 9/10
6/6 ----- 0s 65ms/step - accuracy: 1.0000 - loss: 0.0178 - val_accuracy: 1.0000 - val_loss: 0.0441
Epoch 10/10
6/6 ----- 0s 64ms/step - accuracy: 1.0000 - loss: 0.0156 - val_accuracy: 1.0000 - val_loss: 0.0420
```



	precision	recall	f1-score	support
baseball	1.00	1.00	1.00	3
golf	1.00	1.00	1.00	9
tenis	1.00	1.00	1.00	6
volleyball	1.00	1.00	1.00	10
accuracy			1.00	28
macro avg	1.00	1.00	1.00	28
weighted avg	1.00	1.00	1.00	28

[[3 0 0 0]
[0 9 0 0]
[0 0 6 0]
[0 0 0 10]]



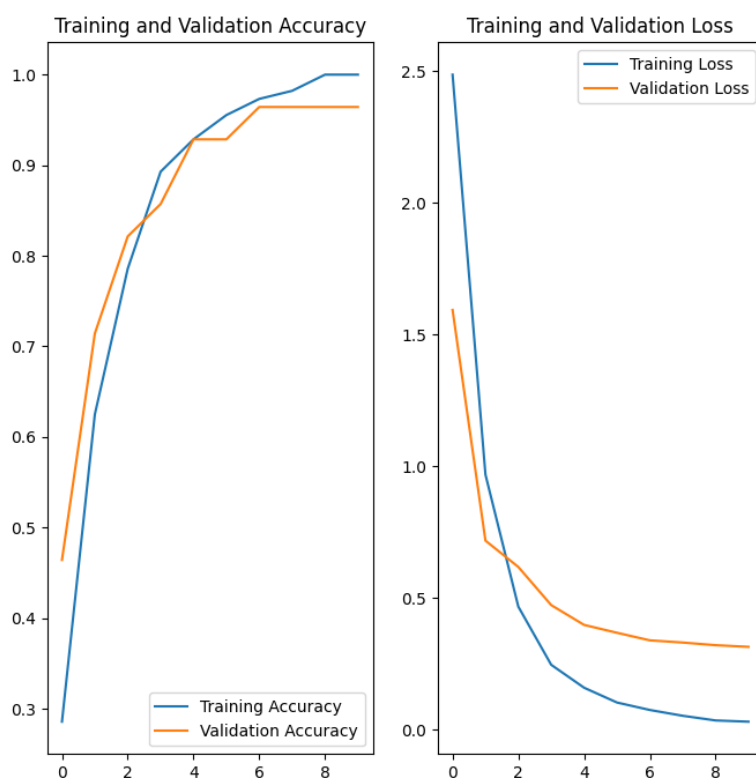
4f. Różny rozmiar Batch size (np. 32, 64 ,128) – raport porównawczy dla każdego rozmiaru

	CZAS	DOKŁADNOŚĆ
Batch size 32	24 sek.	0,96
Batch size 64	28 sek.	0,93
Batch size 128	17 sek.	0,79

Batch size 32:

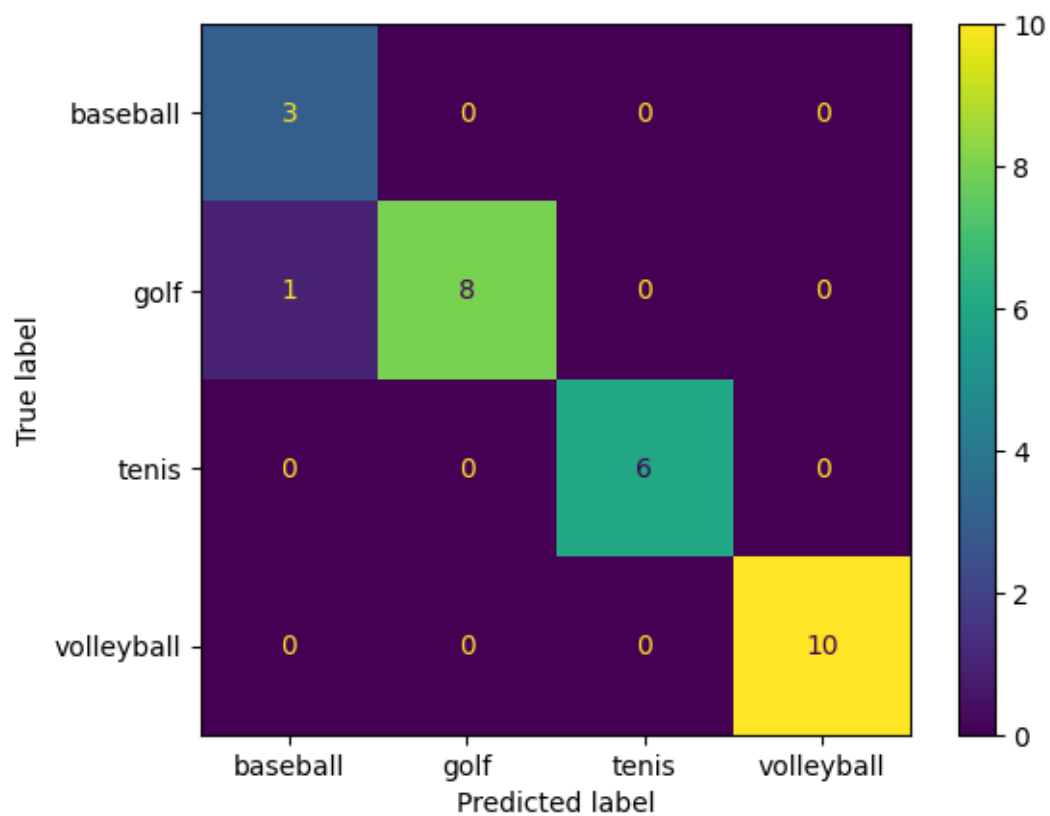
```
[63] epochs=10
history = model.fit(
    train_ds,
    validation_data=val_ds,
    epochs=epochs
)

Epoch 1/10
/usr/local/lib/python3.11/dist-packages/keras/src/backend/tensorflow/nn.py:708: UserWarning: "sparse_categorical_crossentropy"
output, from_logits = _get_logits(
4/4 23s 4s/step - accuracy: 0.2591 - loss: 2.5461 - val_accuracy: 0.4643 - val_loss: 1.5925
Epoch 2/10
4/4 0s 44ms/step - accuracy: 0.5929 - loss: 1.1399 - val_accuracy: 0.7143 - val_loss: 0.7167
Epoch 3/10
4/4 0s 47ms/step - accuracy: 0.7814 - loss: 0.4947 - val_accuracy: 0.8214 - val_loss: 0.6167
Epoch 4/10
4/4 0s 29ms/step - accuracy: 0.8696 - loss: 0.2555 - val_accuracy: 0.8571 - val_loss: 0.4719
Epoch 5/10
4/4 0s 27ms/step - accuracy: 0.9356 - loss: 0.1556 - val_accuracy: 0.9286 - val_loss: 0.3967
Epoch 6/10
4/4 0s 32ms/step - accuracy: 0.9596 - loss: 0.0981 - val_accuracy: 0.9286 - val_loss: 0.3669
Epoch 7/10
4/4 0s 28ms/step - accuracy: 0.9799 - loss: 0.0622 - val_accuracy: 0.9643 - val_loss: 0.3381
Epoch 8/10
4/4 0s 27ms/step - accuracy: 0.9699 - loss: 0.0655 - val_accuracy: 0.9643 - val_loss: 0.3298
Epoch 9/10
4/4 0s 28ms/step - accuracy: 1.0000 - loss: 0.0316 - val_accuracy: 0.9643 - val_loss: 0.3197
Epoch 10/10
4/4 0s 30ms/step - accuracy: 1.0000 - loss: 0.0317 - val_accuracy: 0.9643 - val_loss: 0.3135
```



	precision	recall	f1-score	support
baseball	0.75	1.00	0.86	3
golf	1.00	0.89	0.94	9
tenis	1.00	1.00	1.00	6
volleyball	1.00	1.00	1.00	10
accuracy			0.96	28
macro avg	0.94	0.97	0.95	28
weighted avg	0.97	0.96	0.97	28

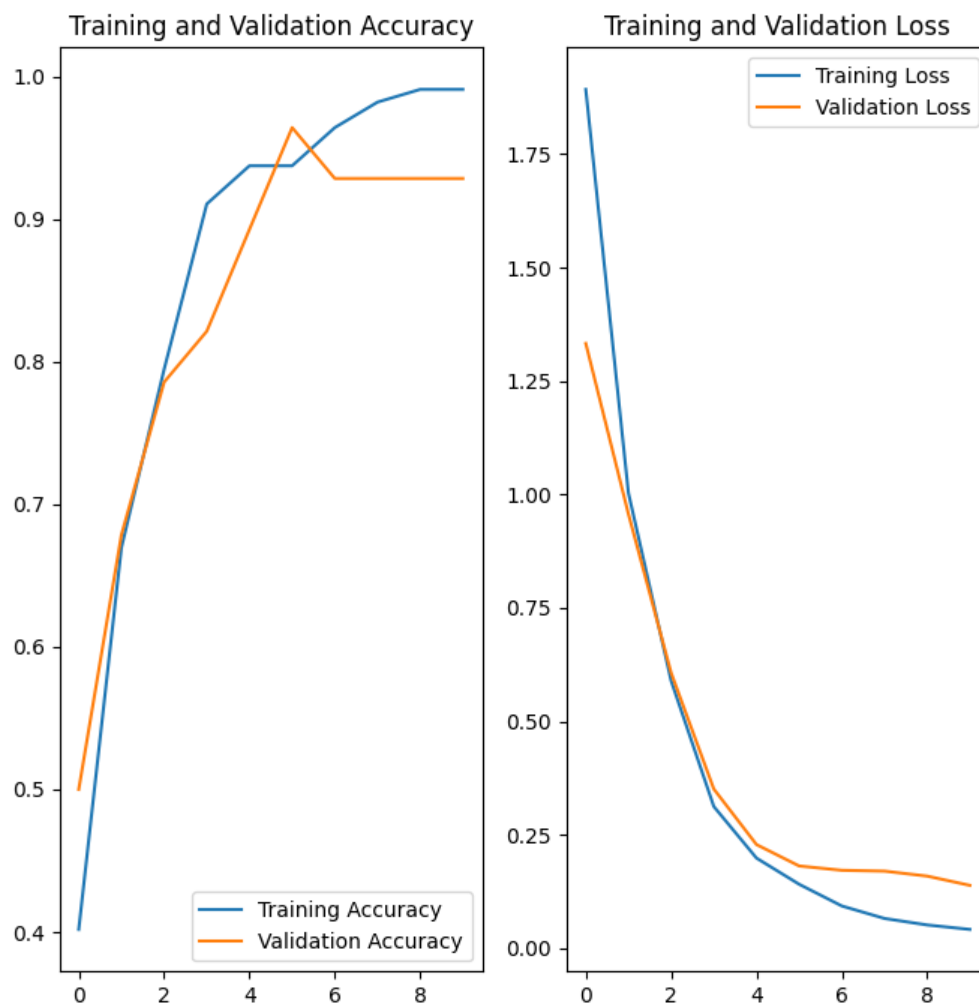

```
[[ 3  0  0  0]
 [ 1  8  0  0]
 [ 0  0  6  0]
 [ 0  0  0 10]]
```



Batch size 64:

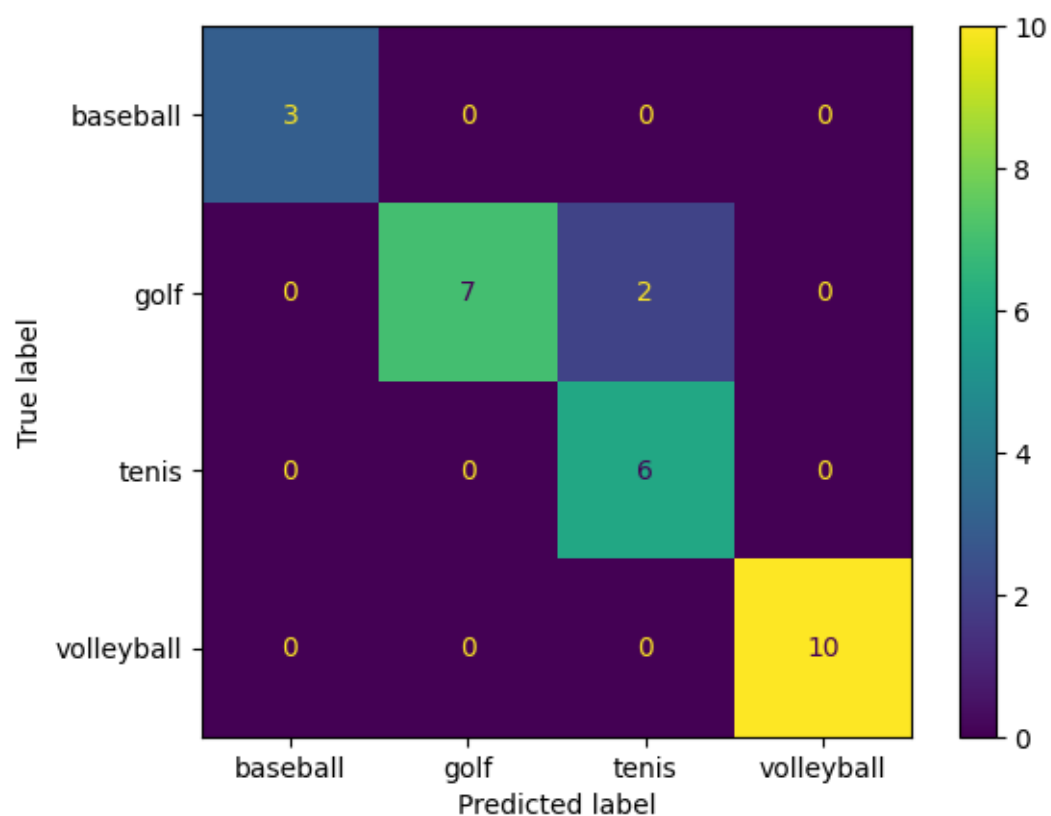
```
[73] epochs=10
history = model.fit(
    train_ds,
    validation_data=val_ds,
    epochs=epochs
)

Epoch 1/10
/usr/local/lib/python3.11/dist-packages/keras/src/backend/tensorflow/nn.py:708: UserWarning: "`sparse_categorical_crossentropy`
output, from_logits = _get_logits(
2/2 ----- 21s 9s/step - accuracy: 0.3512 - loss: 2.1794 - val_accuracy: 0.5000 - val_loss: 1.3329
Epoch 2/10
2/2 ----- 7s 97ms/step - accuracy: 0.6548 - loss: 1.1166 - val_accuracy: 0.6786 - val_loss: 0.9567
Epoch 3/10
2/2 ----- 0s 63ms/step - accuracy: 0.7798 - loss: 0.5873 - val_accuracy: 0.7857 - val_loss: 0.6068
Epoch 4/10
2/2 ----- 0s 67ms/step - accuracy: 0.9058 - loss: 0.3516 - val_accuracy: 0.8214 - val_loss: 0.3508
Epoch 5/10
2/2 ----- 0s 57ms/step - accuracy: 0.9323 - loss: 0.1952 - val_accuracy: 0.8929 - val_loss: 0.2283
Epoch 6/10
2/2 ----- 0s 57ms/step - accuracy: 0.9323 - loss: 0.1399 - val_accuracy: 0.9643 - val_loss: 0.1811
Epoch 7/10
2/2 ----- 0s 57ms/step - accuracy: 0.9658 - loss: 0.0946 - val_accuracy: 0.9286 - val_loss: 0.1716
Epoch 8/10
2/2 ----- 0s 54ms/step - accuracy: 0.9829 - loss: 0.0674 - val_accuracy: 0.9286 - val_loss: 0.1698
Epoch 9/10
2/2 ----- 0s 59ms/step - accuracy: 0.9888 - loss: 0.0529 - val_accuracy: 0.9286 - val_loss: 0.1588
Epoch 10/10
2/2 ----- 0s 55ms/step - accuracy: 0.9888 - loss: 0.0433 - val_accuracy: 0.9286 - val_loss: 0.1384
```



	precision	recall	f1-score	support
baseball	1.00	1.00	1.00	3
golf	1.00	0.78	0.88	9
tenis	0.75	1.00	0.86	6
volleyball	1.00	1.00	1.00	10
accuracy			0.93	28
macro avg	0.94	0.94	0.93	28
weighted avg	0.95	0.93	0.93	28

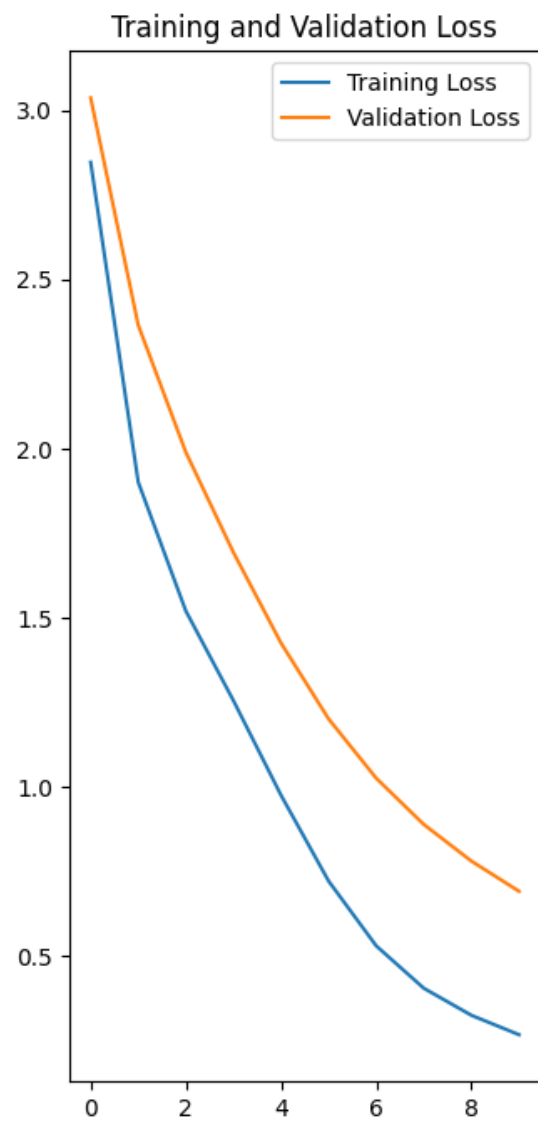
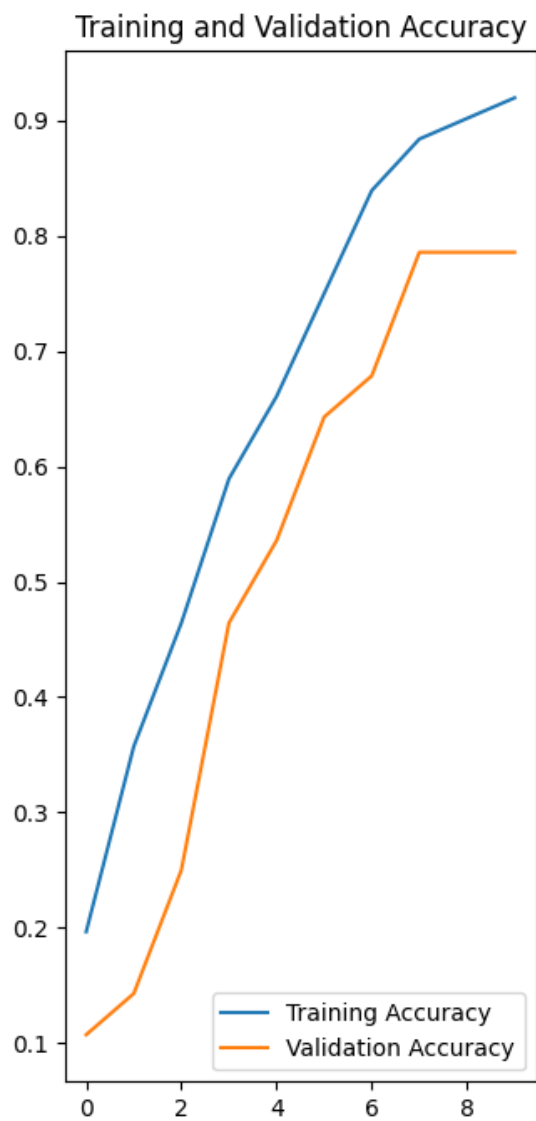

```
[[ 3  0  0  0]
 [ 0  7  2  0]
 [ 0  0  6  0]
 [ 0  0  0 10]]
```



Batch size 128:

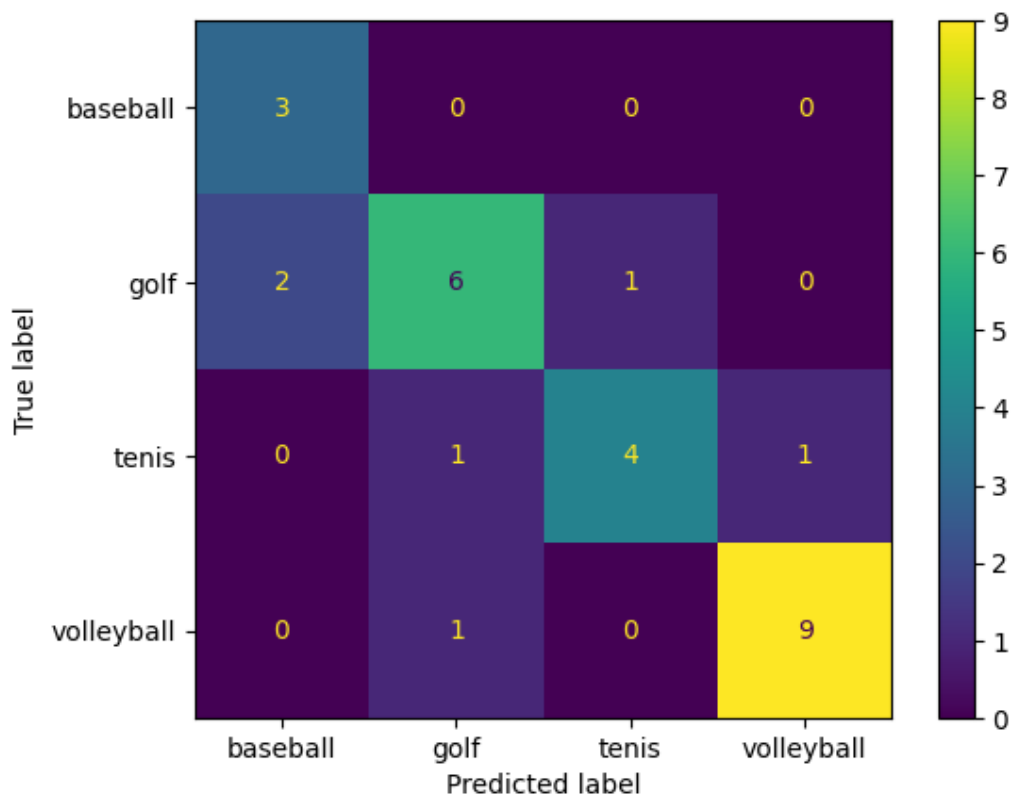
```
[93] epochs=10
history = model.fit(
    train_ds,
    validation_data=val_ds,
    epochs=epochs
)

Epoch 1/10
/usr/local/lib/python3.11/dist-packages/keras/src/backend/tensorflow/nn.py:708: UserWarning: "sparse_categorical_crossentropy"
output, from_logits = _get_logits(
1/1 ----- 15s 15s/step - accuracy: 0.1964 - loss: 2.8467 - val_accuracy: 0.1071 - val_loss: 3.0370
Epoch 2/10
1/1 ----- 2s 2s/step - accuracy: 0.3571 - loss: 1.9012 - val_accuracy: 0.1429 - val_loss: 2.3653
Epoch 3/10
1/1 ----- 0s 109ms/step - accuracy: 0.4643 - loss: 1.5207 - val_accuracy: 0.2500 - val_loss: 1.9897
Epoch 4/10
1/1 ----- 0s 103ms/step - accuracy: 0.5893 - loss: 1.2558 - val_accuracy: 0.4643 - val_loss: 1.6936
Epoch 5/10
1/1 ----- 0s 87ms/step - accuracy: 0.6607 - loss: 0.9761 - val_accuracy: 0.5357 - val_loss: 1.4259
Epoch 6/10
1/1 ----- 0s 171ms/step - accuracy: 0.7500 - loss: 0.7225 - val_accuracy: 0.6429 - val_loss: 1.2017
Epoch 7/10
1/1 ----- 0s 100ms/step - accuracy: 0.8393 - loss: 0.5312 - val_accuracy: 0.6786 - val_loss: 1.0265
Epoch 8/10
1/1 ----- 0s 83ms/step - accuracy: 0.8839 - loss: 0.4059 - val_accuracy: 0.7857 - val_loss: 0.8899
Epoch 9/10
1/1 ----- 0s 82ms/step - accuracy: 0.9018 - loss: 0.3260 - val_accuracy: 0.7857 - val_loss: 0.7817
Epoch 10/10
1/1 ----- 0s 79ms/step - accuracy: 0.9196 - loss: 0.2687 - val_accuracy: 0.7857 - val_loss: 0.6925
```



	precision	recall	f1-score	support
baseball	0.60	1.00	0.75	3
golf	0.75	0.67	0.71	9
tenis	0.80	0.67	0.73	6
volleyball	0.90	0.90	0.90	10
accuracy			0.79	28
macro avg	0.76	0.81	0.77	28
weighted avg	0.80	0.79	0.78	28


```
[[3 0 0 0]
 [2 6 1 0]
 [0 1 4 1]
 [0 1 0 9]]
```



4g. Różne struktury sieci (np.. VGG16, ResNet101, InceptionV3, MobileNet) – raport porównawczy dla każdej sieci

	CZAS	DOKŁADNOŚĆ
VGG16	52 sek.	0,93
ResNet101	50 sek.	1,00
InceptionV3	54 sek.	0,50
MobileNet	26 sek.	0,61

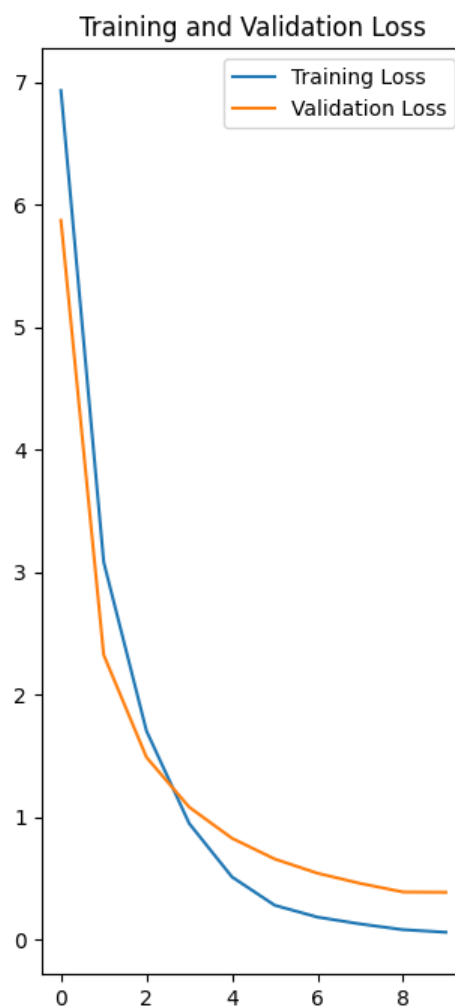
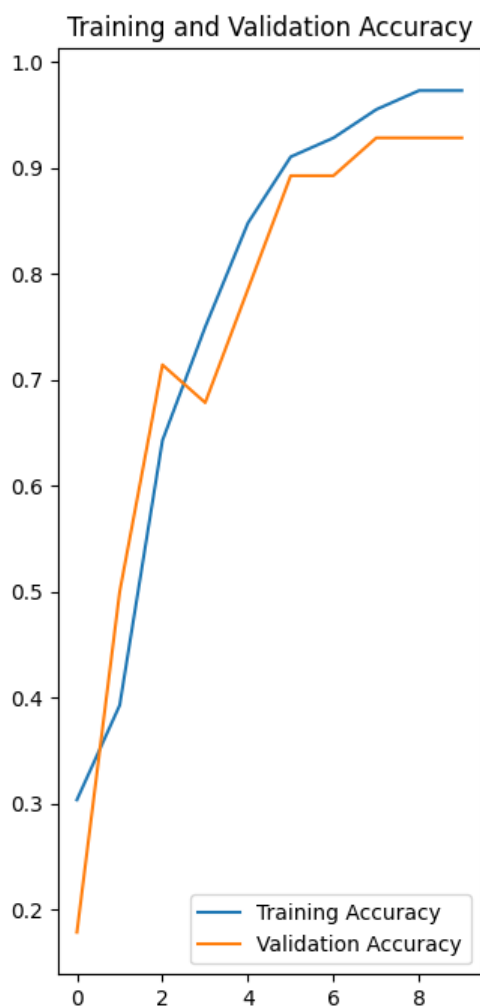
Dla każdego modelu używam takich danych wejściowych:

```
batch_size = 20
class_count = 4

img_height = 224
img_width = 224
```

VGG16:

```
Epoch 1/10
/usr/local/lib/python3.11/dist-packages/keras/src/backend/tensorflow/nn.py:708: UserWarning: "`sparse_categorical_crossentropy"
  output, from_logits = _get_logits(
6/6 ----- 26s 3s/step - accuracy: 0.3525 - loss: 7.0346 - val_accuracy: 0.1786 - val_loss: 5.8703
Epoch 2/10
6/6 ----- 17s 147ms/step - accuracy: 0.3319 - loss: 4.0226 - val_accuracy: 0.5000 - val_loss: 2.3231
Epoch 3/10
6/6 ----- 1s 148ms/step - accuracy: 0.6940 - loss: 1.5890 - val_accuracy: 0.7143 - val_loss: 1.4875
Epoch 4/10
6/6 ----- 1s 149ms/step - accuracy: 0.7295 - loss: 0.9889 - val_accuracy: 0.6786 - val_loss: 1.0803
Epoch 5/10
6/6 ----- 1s 150ms/step - accuracy: 0.7948 - loss: 0.6905 - val_accuracy: 0.7857 - val_loss: 0.8264
Epoch 6/10
6/6 ----- 1s 150ms/step - accuracy: 0.9196 - loss: 0.3231 - val_accuracy: 0.8929 - val_loss: 0.6569
Epoch 7/10
6/6 ----- 1s 150ms/step - accuracy: 0.9032 - loss: 0.1890 - val_accuracy: 0.8929 - val_loss: 0.5408
Epoch 8/10
6/6 ----- 1s 151ms/step - accuracy: 0.9259 - loss: 0.1650 - val_accuracy: 0.9286 - val_loss: 0.4576
Epoch 9/10
6/6 ----- 1s 161ms/step - accuracy: 0.9861 - loss: 0.0627 - val_accuracy: 0.9286 - val_loss: 0.3876
Epoch 10/10
6/6 ----- 1s 149ms/step - accuracy: 0.9550 - loss: 0.0756 - val_accuracy: 0.9286 - val_loss: 0.3857
```

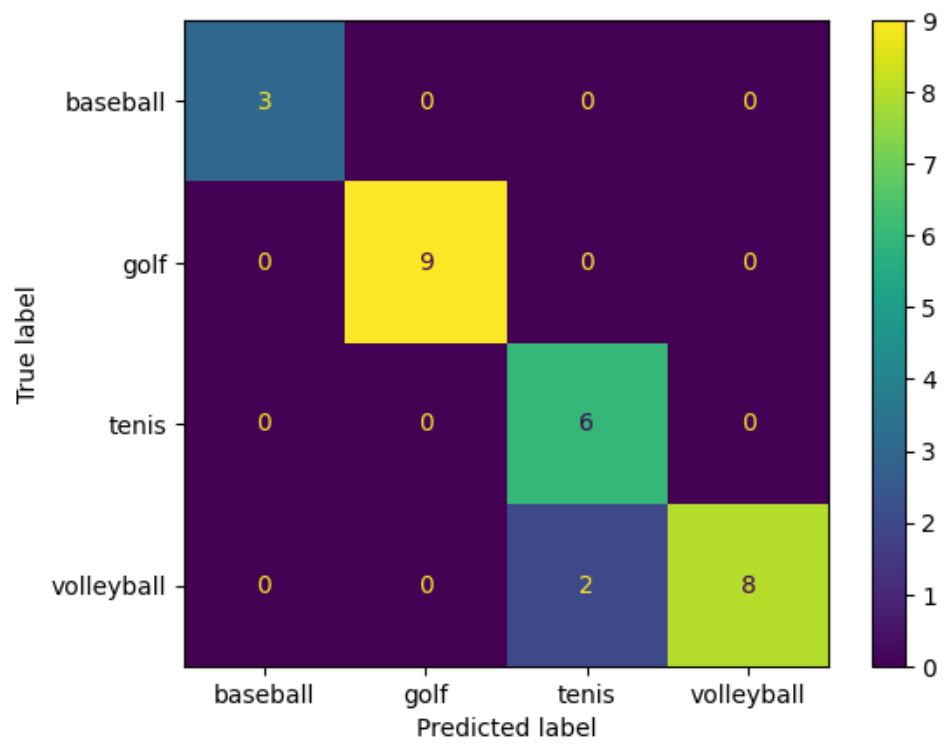


	precision	recall	f1-score	support
baseball	1.00	1.00	1.00	3
golf	1.00	1.00	1.00	9
tenis	0.75	1.00	0.86	6
volleyball	1.00	0.80	0.89	10
accuracy			0.93	28
macro avg	0.94	0.95	0.94	28
weighted avg	0.95	0.93	0.93	28

```

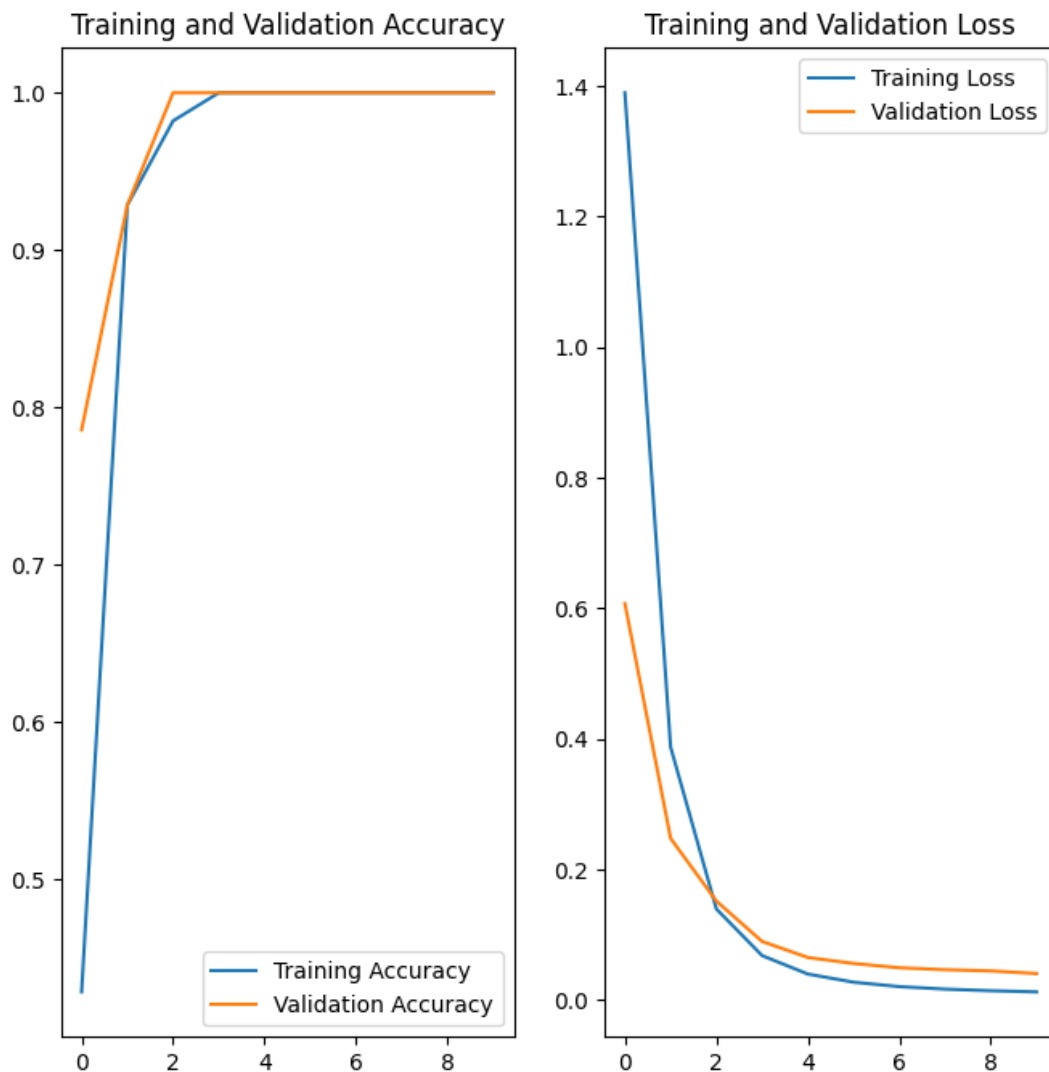
[[3 0 0 0]
 [0 9 0 0]
 [0 0 6 0]
 [0 0 2 8]]

```



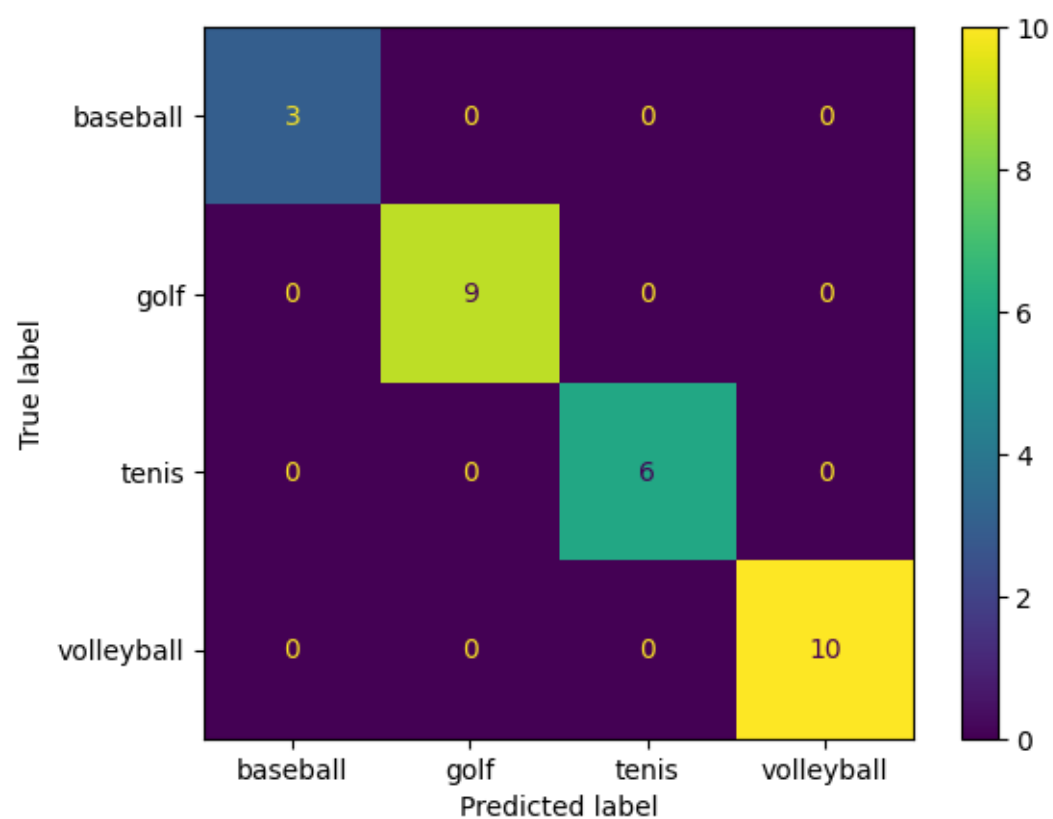
ResNet101:

```
Epoch 1/10  
/usr/local/lib/python3.11/dist-packages/keras/src/backend/tensorflow/nn.py:708: UserWarning: "`sparse_categorical_crossentropy, from_logits = _get_logits(  
6/6 ————— 41s 4s/step - accuracy: 0.3421 - loss: 1.5992 - val_accuracy: 0.7857 - val_loss: 0.6073  
Epoch 2/10  
6/6 ————— 1s 123ms/step - accuracy: 0.9164 - loss: 0.4349 - val_accuracy: 0.9286 - val_loss: 0.2478  
Epoch 3/10  
6/6 ————— 1s 118ms/step - accuracy: 0.9898 - loss: 0.1483 - val_accuracy: 1.0000 - val_loss: 0.1519  
Epoch 4/10  
6/6 ————— 1s 119ms/step - accuracy: 1.0000 - loss: 0.0754 - val_accuracy: 1.0000 - val_loss: 0.0900  
Epoch 5/10  
6/6 ————— 1s 124ms/step - accuracy: 1.0000 - loss: 0.0472 - val_accuracy: 1.0000 - val_loss: 0.0655  
Epoch 6/10  
6/6 ————— 1s 119ms/step - accuracy: 1.0000 - loss: 0.0281 - val_accuracy: 1.0000 - val_loss: 0.0563  
Epoch 7/10  
6/6 ————— 1s 118ms/step - accuracy: 1.0000 - loss: 0.0229 - val_accuracy: 1.0000 - val_loss: 0.0498  
Epoch 8/10  
6/6 ————— 1s 119ms/step - accuracy: 1.0000 - loss: 0.0181 - val_accuracy: 1.0000 - val_loss: 0.0468  
Epoch 9/10  
6/6 ————— 1s 118ms/step - accuracy: 1.0000 - loss: 0.0157 - val_accuracy: 1.0000 - val_loss: 0.0449  
Epoch 10/10  
6/6 ————— 1s 126ms/step - accuracy: 1.0000 - loss: 0.0131 - val_accuracy: 1.0000 - val_loss: 0.0409
```



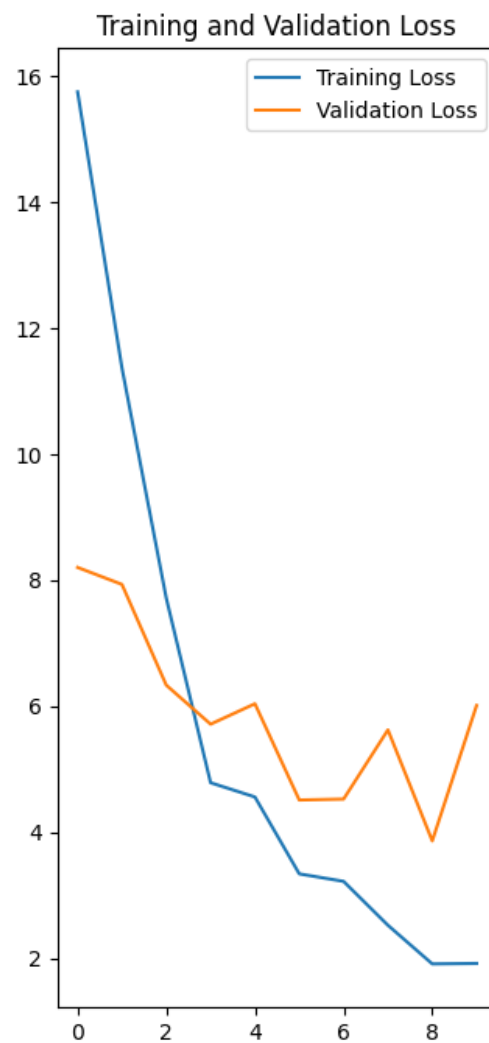
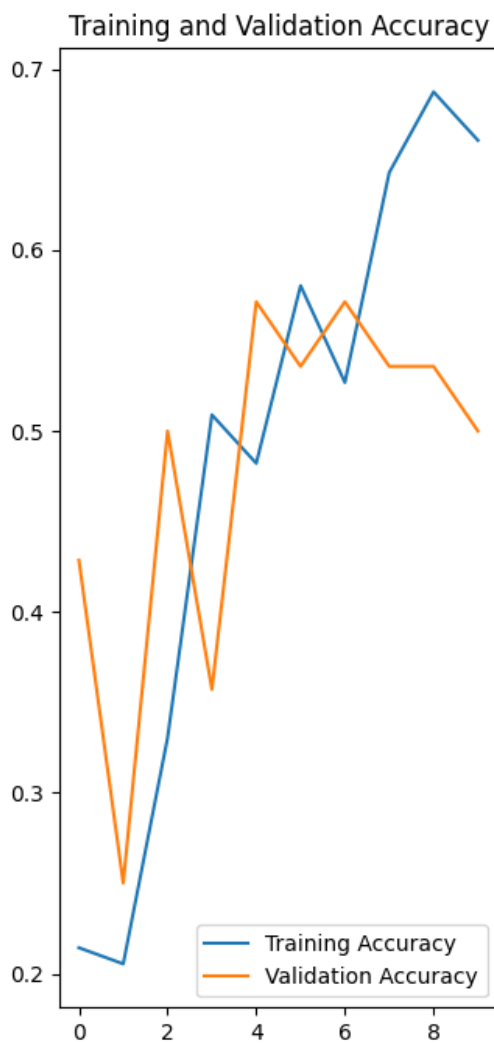
	precision	recall	f1-score	support
baseball	1.00	1.00	1.00	3
golf	1.00	1.00	1.00	9
tenis	1.00	1.00	1.00	6
volleyball	1.00	1.00	1.00	10
accuracy			1.00	28
macro avg	1.00	1.00	1.00	28
weighted avg	1.00	1.00	1.00	28


```
[[ 3  0  0  0]
 [ 0  9  0  0]
 [ 0  0  6  0]
 [ 0  0  0 10]]
```



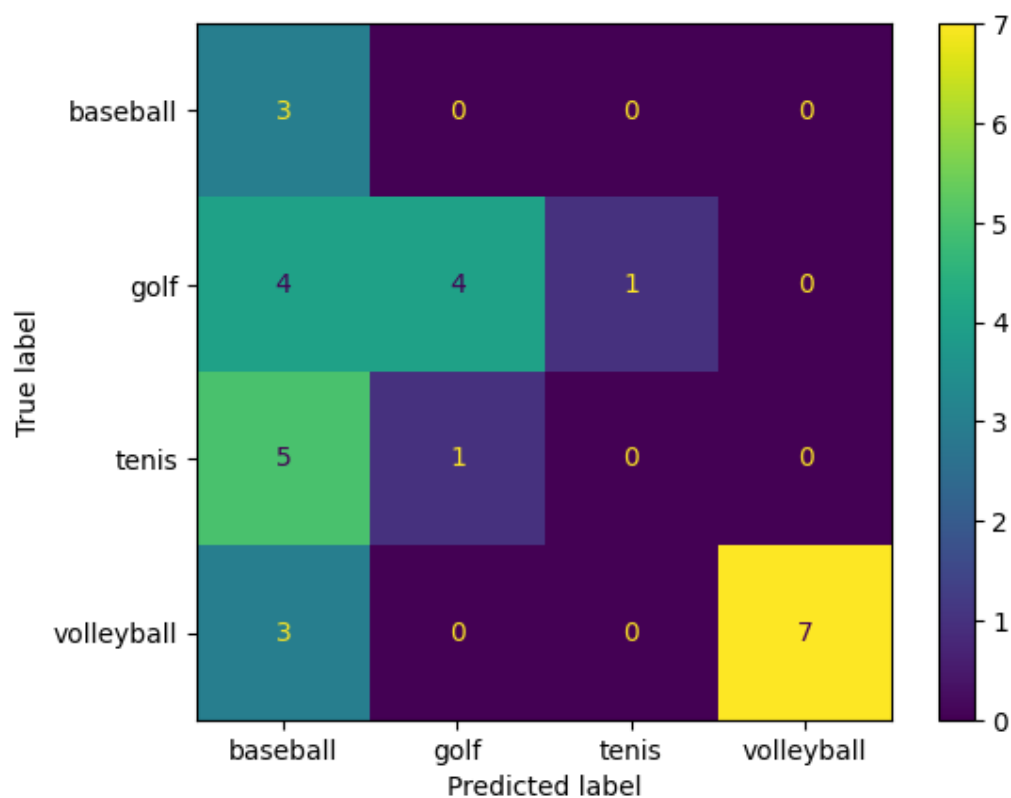
InceptionV3:

```
Epoch 1/10
/usr/local/lib/python3.11/dist-packages/keras/src/backend/tensorflow/nn.py:708: UserWarning: ""sparse_categorical_crossentropy"
output, from_logits = _get_logits(
6/6 ----- 32s 3s/step - accuracy: 0.2046 - loss: 13.8734 - val_accuracy: 0.4286 - val_loss: 8.2031
Epoch 2/10
6/6 ----- 19s 65ms/step - accuracy: 0.2266 - loss: 10.1217 - val_accuracy: 0.2500 - val_loss: 7.9336
Epoch 3/10
6/6 ----- 1s 70ms/step - accuracy: 0.3280 - loss: 8.0489 - val_accuracy: 0.5000 - val_loss: 6.3352
Epoch 4/10
6/6 ----- 0s 54ms/step - accuracy: 0.4766 - loss: 4.9757 - val_accuracy: 0.3571 - val_loss: 5.7159
Epoch 5/10
6/6 ----- 0s 56ms/step - accuracy: 0.4669 - loss: 4.8014 - val_accuracy: 0.5714 - val_loss: 6.0399
Epoch 6/10
6/6 ----- 1s 56ms/step - accuracy: 0.5413 - loss: 3.6891 - val_accuracy: 0.5357 - val_loss: 4.5127
Epoch 7/10
6/6 ----- 1s 57ms/step - accuracy: 0.5286 - loss: 3.7964 - val_accuracy: 0.5714 - val_loss: 4.5273
Epoch 8/10
6/6 ----- 0s 72ms/step - accuracy: 0.6520 - loss: 2.3169 - val_accuracy: 0.5357 - val_loss: 5.6253
Epoch 9/10
6/6 ----- 0s 53ms/step - accuracy: 0.6772 - loss: 2.0443 - val_accuracy: 0.5357 - val_loss: 3.8639
Epoch 10/10
6/6 ----- 0s 54ms/step - accuracy: 0.6680 - loss: 1.7762 - val_accuracy: 0.5000 - val_loss: 6.0126
```



	precision	recall	f1-score	support
baseball	0.20	1.00	0.33	3
golf	0.80	0.44	0.57	9
tenis	0.00	0.00	0.00	6
volleyball	1.00	0.70	0.82	10
accuracy			0.50	28
macro avg	0.50	0.54	0.43	28
weighted avg	0.64	0.50	0.51	28

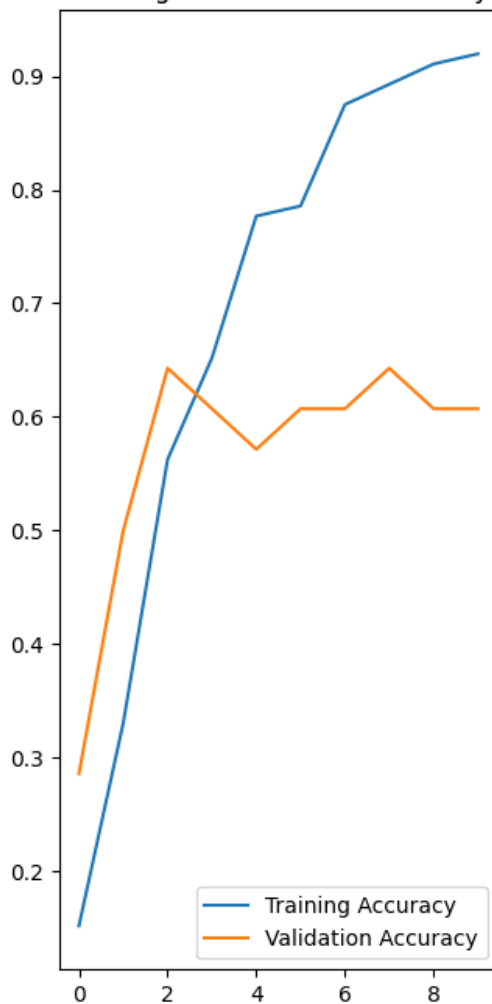

```
[[3 0 0 0]
 [4 4 1 0]
 [5 1 0 0]
 [3 0 0 7]]
```



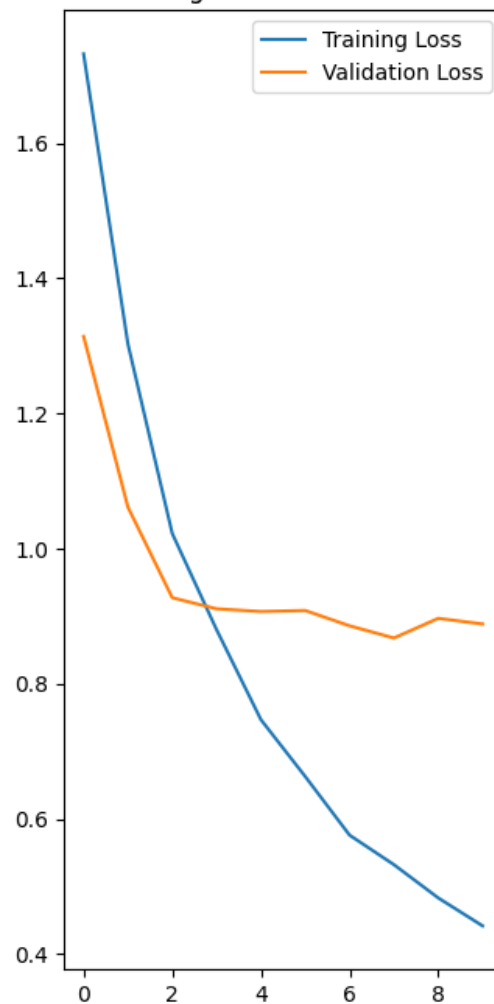
MobileNet:

```
Epoch 1/10  
/usr/local/lib/python3.11/dist-packages/keras/src/backend/tensorflow/nn.py:708: UserWarning: "`sparse_categorical_crossentropy` does not support the `from_logits` argument. To avoid this warning, please use `sparse_categorical_crossentropy` without the `from_logits` argument, or consider using `sparse_categorical_crossentropy` with `from_logits=True`."`  
  output, from_logits = _get_logits(  
6/6 ----- 16s 2s/step - accuracy: 0.1218 - loss: 1.8734 - val_accuracy: 0.2857 - val_loss: 1.3142  
Epoch 2/10  
6/6 ----- 10s 39ms/step - accuracy: 0.2744 - loss: 1.3823 - val_accuracy: 0.5000 - val_loss: 1.0615  
Epoch 3/10  
6/6 ----- 0s 30ms/step - accuracy: 0.5654 - loss: 1.0482 - val_accuracy: 0.6429 - val_loss: 0.9274  
Epoch 4/10  
6/6 ----- 0s 33ms/step - accuracy: 0.6746 - loss: 0.8799 - val_accuracy: 0.6071 - val_loss: 0.9110  
Epoch 5/10  
6/6 ----- 0s 27ms/step - accuracy: 0.7766 - loss: 0.7402 - val_accuracy: 0.5714 - val_loss: 0.9070  
Epoch 6/10  
6/6 ----- 0s 27ms/step - accuracy: 0.8113 - loss: 0.6693 - val_accuracy: 0.6071 - val_loss: 0.9085  
Epoch 7/10  
6/6 ----- 0s 32ms/step - accuracy: 0.8605 - loss: 0.5823 - val_accuracy: 0.6071 - val_loss: 0.8858  
Epoch 8/10  
6/6 ----- 0s 26ms/step - accuracy: 0.8632 - loss: 0.5496 - val_accuracy: 0.6429 - val_loss: 0.8678  
Epoch 9/10  
6/6 ----- 0s 22ms/step - accuracy: 0.9148 - loss: 0.5175 - val_accuracy: 0.6071 - val_loss: 0.8969  
Epoch 10/10  
6/6 ----- 0s 23ms/step - accuracy: 0.9186 - loss: 0.4331 - val_accuracy: 0.6071 - val_loss: 0.8886
```

Training and Validation Accuracy



Training and Validation Loss



	precision	recall	f1-score	support
baseball	0.25	0.33	0.29	3
golf	0.50	0.22	0.31	9
tenis	0.45	0.83	0.59	6
volleyball	1.00	0.90	0.95	10
accuracy			0.61	28
macro avg	0.55	0.57	0.53	28
weighted avg	0.64	0.61	0.59	28

```

[[1 0 2 0]
 [3 2 4 0]
 [0 1 5 0]
 [0 1 0 9]]

```

